

SMRs without water the gas-cooled reactor

Helium, graphite and TRISO, then reading **HTR-PM** component by component

Week 5 · Instructor **Minseop Song** · Rev1.0 · September 2026

How the textbook is organised - **four parts**, three decks

- Each reactor gets five chapters: **principle** → **fuel and materials** → **systems** → **safety** → **system code**. The system-code chapter is what sets this book apart
- Volume III, ch. 6 of *Writing a Reactor System Code* said **swap the property class** and the code runs another fluid - this book actually does it

1**Part 1 · VHTR / HTGR**

coolant helium · chapters 1~6

How do you cool a reactor with a gas that barely carries heat?

this deck · week 5

2**Part 2 · SFR**

coolant sodium · chapters 7~11

Why use a metal that explodes on contact with water?

next deck · weeks 5-6

3**Part 3 · MSR**

coolant molten salt · chapters 12~16

What does it mean for the fuel to flow?

third deck · week 6

4**Part 4 · synthesis**

coolant — · chapters 17~18

Measure all three with one ruler - what shows?

end of the third deck

01 Why give up water

Every SMR so far removed heat with water and slowed neutrons with water. The reason to give it up comes from one line of the property table. Only once you also count what the alternatives bring does “safer” mean anything.

Four **constraints of water**, and four new problems without it

- Water boils at 100 °C. Holding it liquid at 300 °C takes **15 MPa**, and that one pressure sets nearly the whole design
- Drop water and the four on the left vanish while the four on the right appear - “safer” can only be said **after counting both lists**

WHAT WATER BRINGS

A thick pressure vessel

Only a handful of forges worldwide can make one

LOCA dominates the design basis

ECCS and containment grow to match

325 °C outlet, about 33 % efficiency

Poor steam conditions cap the efficiency

Hydrogen from zirconium and water

The hydrogen that blew apart Fukushima

WHAT GIVING IT UP BRINGS

Helium carries heat poorly

7 MPa, a 4.5 MW blower (Part 1)

Sodium explodes with water

Intermediate loop, steam generator (Part 2)

Salt freezes below 459 °C

Freezing and plugging become accidents (Part 3)

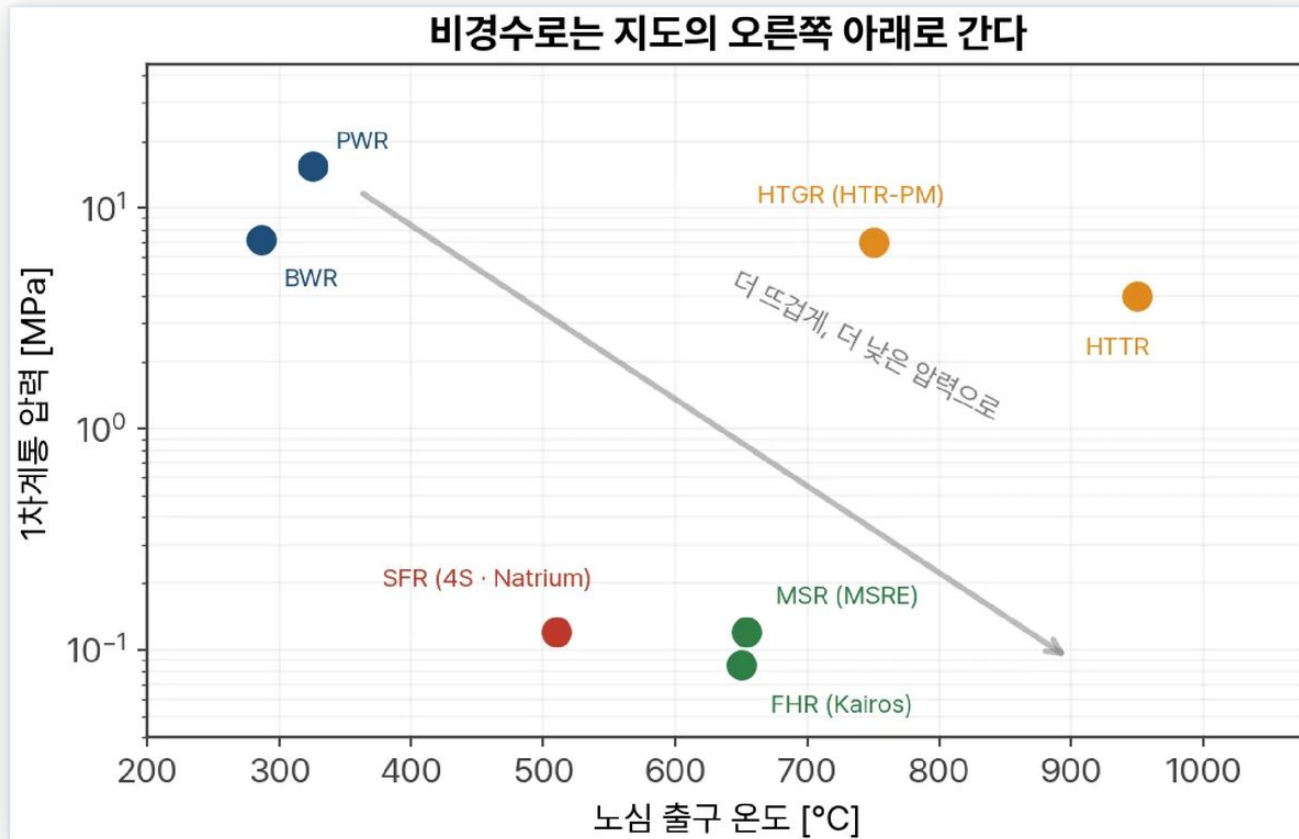
Very little operating experience

Validation data 30 to 60 years old (Part 4)



Core outlet temperature and primary pressure by reactor type

- Every non-LWR sits **lower right** - hotter, and at lower pressure



Core outlet temperature (horizontal) against primary pressure (vertical, log) - textbook Fig. 1.1

LIGHT WATER · PWR

~15 MPa · 325 °C

GAS-COOLED · HTR-PM

7.0 MPa · 750 °C

SODIUM · MOLTEN SALT

~0.1 MPa · 500~700 °C

Near atmospheric pressure **the shape of a LOCA changes**. Nothing flashes to steam and blows out the way water does.

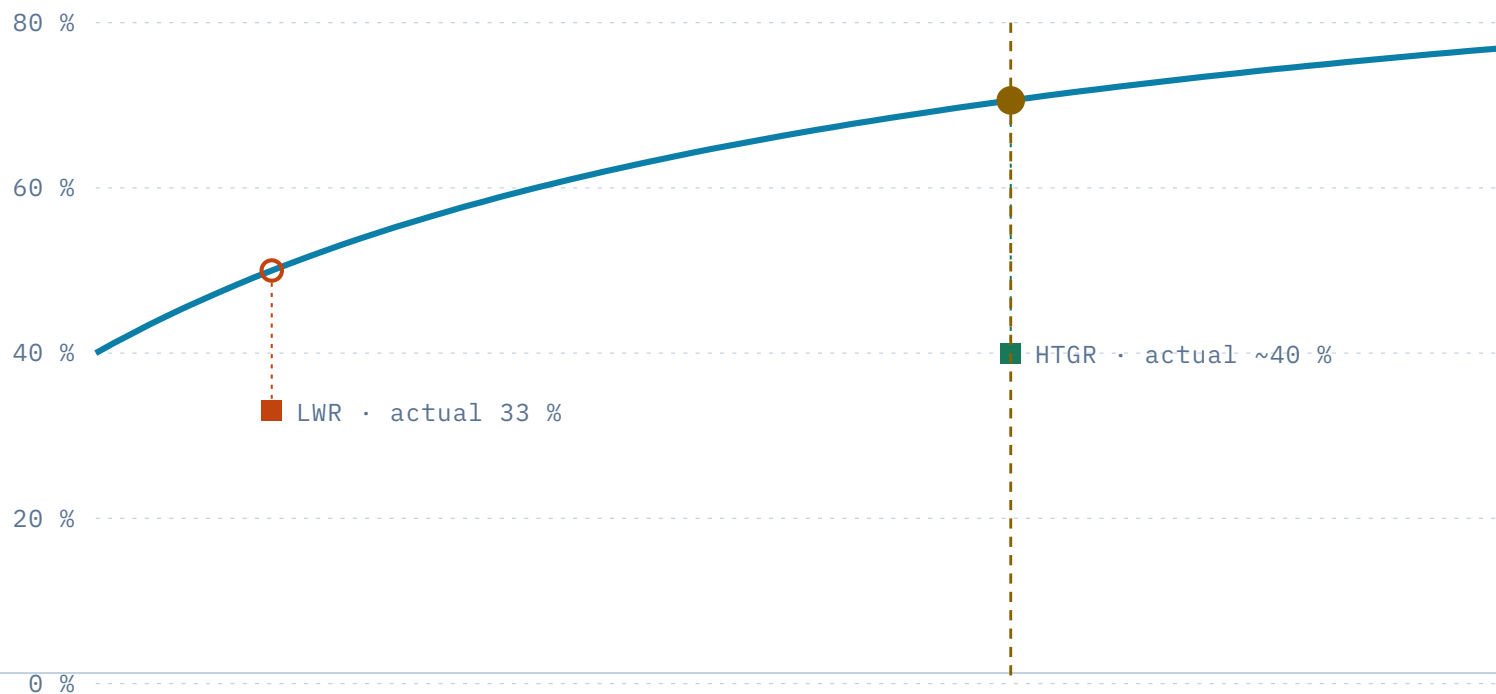
HTGR High-Temperature Gas-cooled Reactor · HTTR Japan's High Temperature engineering Test Reactor · SFR Sodium-cooled Fast Reactor · MSR Molten Salt Reactor · MSRE the 1960s ORNL Molten Salt Reactor Experiment · FHR Fluoride-salt-cooled High-temperature Reactor

Source This textbook §1.1, Fig. 1.1 · the vertical axis is logarithmic

The Carnot limit and pressure-vessel wall thickness

- Raising temperature is **economics** in itself - an LWR tops out at 50 % (33 % actual), an HTGR at 71 % (low 40s actual)
- Wall thickness scales with pressure. From 15.5 to 0.1 MPa it drops to **nearly 1/150** - why SFR and MSR vessels are thin plate, and why "built in a shipyard" is said at all

$$\eta_{\max} = 1 - T_c/T_h \quad t = p R / (\sigma_{\text{allow}} - 0.6 p)$$

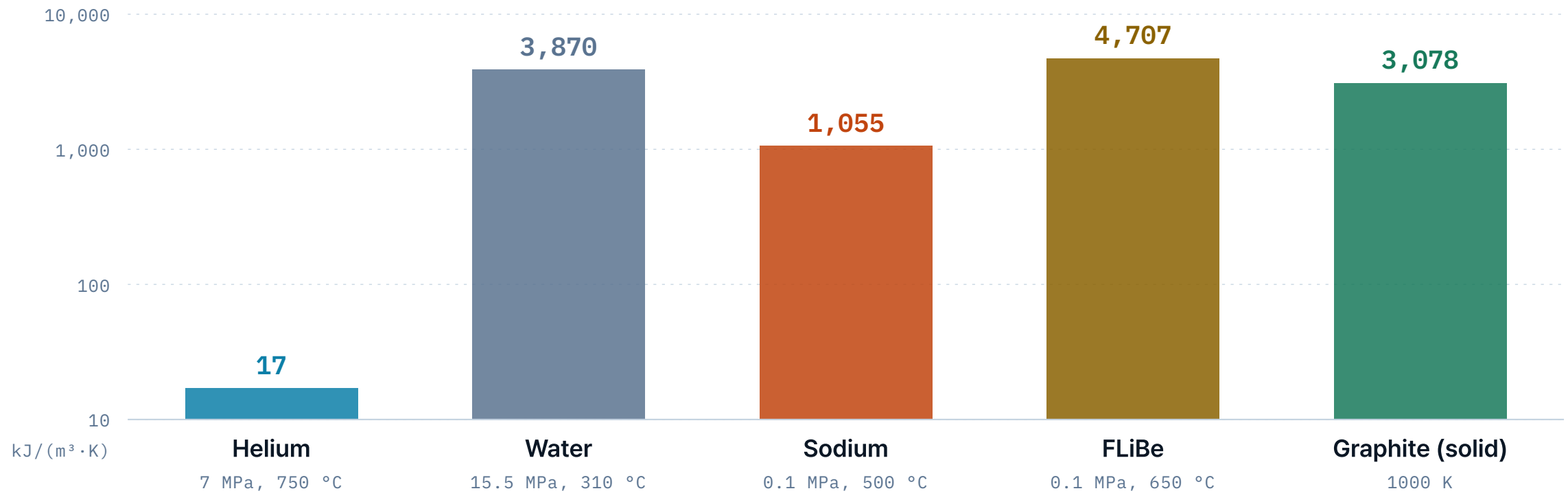


Carnot efficiency the theoretical upper limit for a heat engine between two temperatures · **allowable stress** σ_{allow} the highest stress the design code permits in the material

Source This textbook §1.2 · $T_c = 300$ K · actual efficiency 33 % for LWRs, low 40s for HTGRs · wall from the thin-shell approximation

Volumetric heat capacity ρc_p by coolant

- Helium holds **about 1/225** of water and **about 1/180** of graphite. Hence 7 MPa and a 4.5 MW blower
- But the long graphite bar is why **the core survives for days with the coolant gone** - all of Part 1 is in this one chart



volumetric heat capacity ρc_p heat stored per unit volume per kelvin - the smaller it is, the more volume must be pushed, faster, to carry the same heat · **FLiBe** the LiF-BeF₂ fluoride salt

Source This textbook §1.4, Fig. 1.2 · units $\text{kJ}/(\text{m}^3 \cdot \text{K})$ · logarithmic vertical axis · graphite is a solid

Terms - HTGR, VHTR, FHR and MSR

- HTR-PM runs at 750 °C, so strictly it is an **HTGR**. VHTR is the Generation IV label for 900 °C and up
- **Do not confuse an FHR with a liquid-fuel MSR** - both use fluoride salt, but the FHR’s fuel is solid TRISO

TERM	EXPANSION	MEANING HERE
HTGR	High Temperature Gas-cooled Reactor	All helium-cooled, graphite-moderated reactors
VHTR	Very High Temperature Reactor	Generation IV class aiming at 900 °C and above
FHR	Fluoride-salt-cooled High-temp. Reactor	Solid TRISO fuel, fluoride salt coolant only (Kairos)
MSR	Molten Salt Reactor	Fuel dissolved in the salt, flowing with it (Part 3)
SFR	Sodium-cooled Fast Reactor	Sodium-cooled fast-spectrum reactor (Part 2)

Where THINK Lab’s projects meet this book

- ① **Floating MSR** - in July 2026 the KAERI-KRISO-Samsung Heavy molten-salt container ship got ABS Approval in Principle. Ch. 14
- ② **GAMMA+** - built to solve HTGR air ingress. Ch. 6 explains why
- ③ **Scaling and flow visualisation** - sodium and salt must be replaced by water or surrogate fluids in similarity tests

TRISO tri-structural isotropic coated fuel particle · **AiP** Approval in Principle from a classification society · **ABS** American Bureau of Shipping · **GAMMA+** KAERI’s multi-component flow code for HTGRs

Source This textbook §§1.5-1.6 · usage varies between sources; this book fixes it as below

02 Helium and graphite

As a coolant helium is poor in almost every way. It is used because helium does nothing at all. What it cannot carry, graphite stores; while graphite stores it, TRISO holds on to the fission products.

Why helium, and the **three pillars** of the HTGR

- Every one of helium's merits is **something it does not do** - it does not react, absorb, boil or freeze
- **Remove any one and the logic collapses** - without graphite the heat has nowhere to go, without TRISO there is no containment

Chemically inert

reacts with nothing

Almost no absorption

void reactivity ≈ 0

No phase change

no steam tables, no flow regimes

A gas at any temperature

the limit is set by materials

Helium

carries the heat

leaves what it cannot carry to graphite

Graphite

moderates and stores heat

solid well past 3000 °C

TRISO

holds the fission products

each particle a containment (ch. 3)

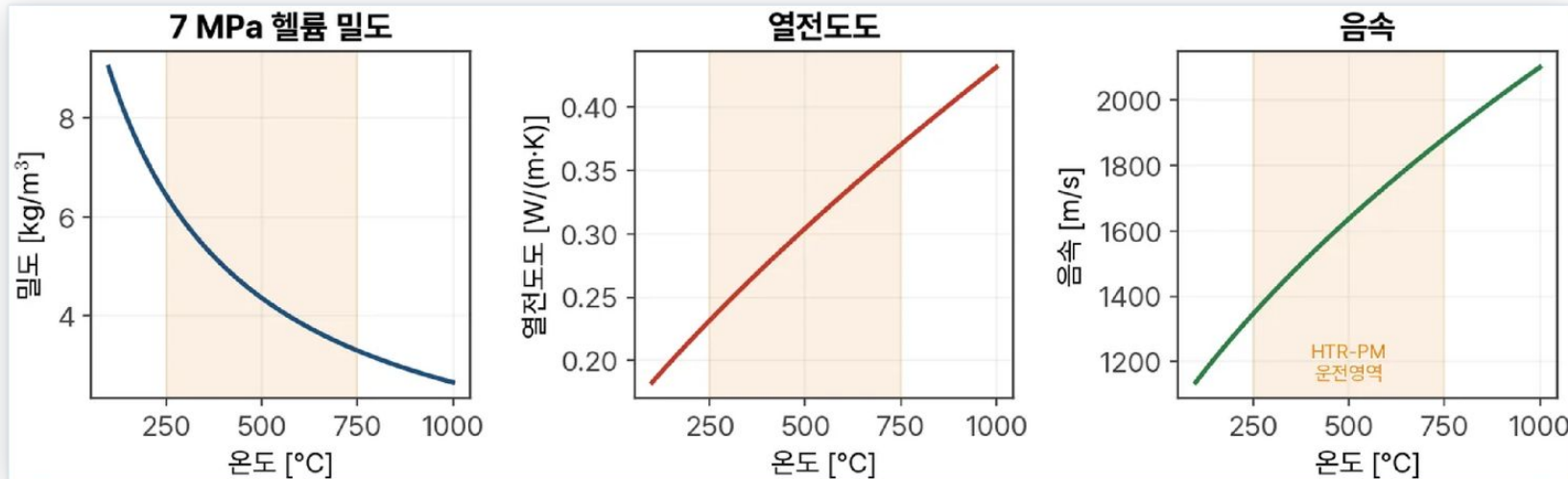
what helium cannot carry, graphite stores; while graphite stores it, TRISO holds the fission products

void reactivity the reactivity change when coolant leaves and voids form · **moderation** slowing neutrons down · **two-phase flow** liquid and gas flowing together

Source This textbook §2.1 · graphite holds about 180 times helium's heat per volume (Fig. 1.2)

Helium properties - ideal gas, constant c_p , the Risø correlations

- Compressibility is **about 1.02** at 7 MPa and 750 °C - ideal to within 2 %. c_p is essentially constant
- The exponents in μ and k are 0.70 and 0.71 $\rightarrow$ **Pr stays near 0.66**. In helium analysis Pr is effectively a constant



Density, thermal conductivity and speed of sound of helium at 7 MPa - textbook Fig. 2.1

$$P = \rho RT, \quad R = 2077 \text{ J/(kg} \cdot \text{K)}$$

$$c_p = 5195 \text{ J/(kg} \cdot \text{K)}, \quad \gamma = 1.667$$

$$\mu = 3.674 \times 10^{-7} T^{0.7} \text{ Pa} \cdot \text{s}$$

$$k = 2.682 \times 10^{-3} (1 + 1.123 \times 10^{-3} P) \times T^{0.71 (1 - 2 \times 10^{-4} P)} \text{ W/(m} \cdot \text{K)}$$

PRANDTL NUMBER PR

$$\approx 0.66 \text{ any } T$$

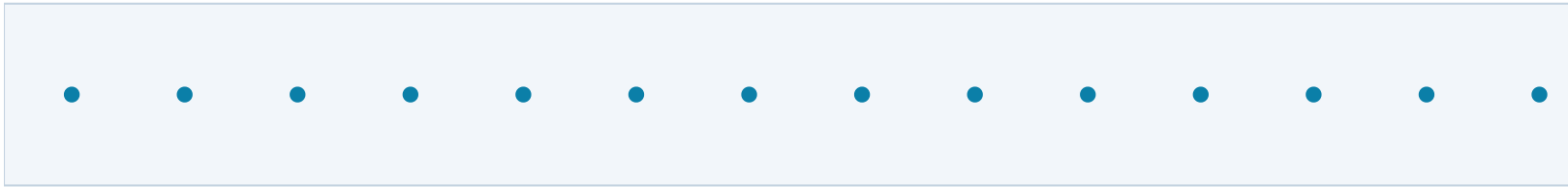
compressibility Z how far a real gas departs from ideal (1 = ideal) · **Prandtl number Pr** ratio of momentum to thermal diffusivity · γ the heat-capacity ratio c_p/c_v

Source This textbook §2.2, Fig. 2.1 · viscosity and conductivity from the Risø correlations (T in K, P in bar) · shading is the HTR-PM range, 250-750 °C

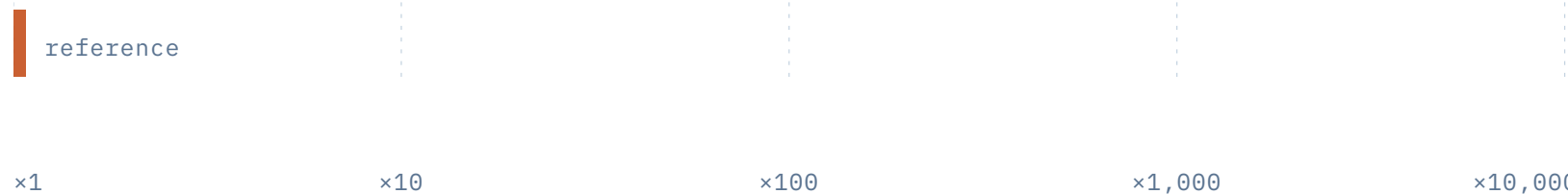
Why helium is held at 7 MPa - blower power

- Even at 7 MPa the volume flow is **about 15 m³/s** - a 3400 MWt LWR pushes 26 m³/s, so this is **eight times more per MW**
- At atmospheric pressure the volume is 70 times and the power $70^2 \approx$ **4900 times**. The pressure is chosen **to make blower power affordable**, not for thermodynamics

One HTR-PM module · 250 MWt · 96.2 kg/s · 250 °C inlet



Blower power multiple (log) - 7 MPa = 1



HELIUM DENSITY ρ

6.44 kg/m³

VOLUME FLOW $\dot{V}$

14.9 m³/s

BLOWER POWER MULTIPLE

×1.0

PRIMARY PRESSURE

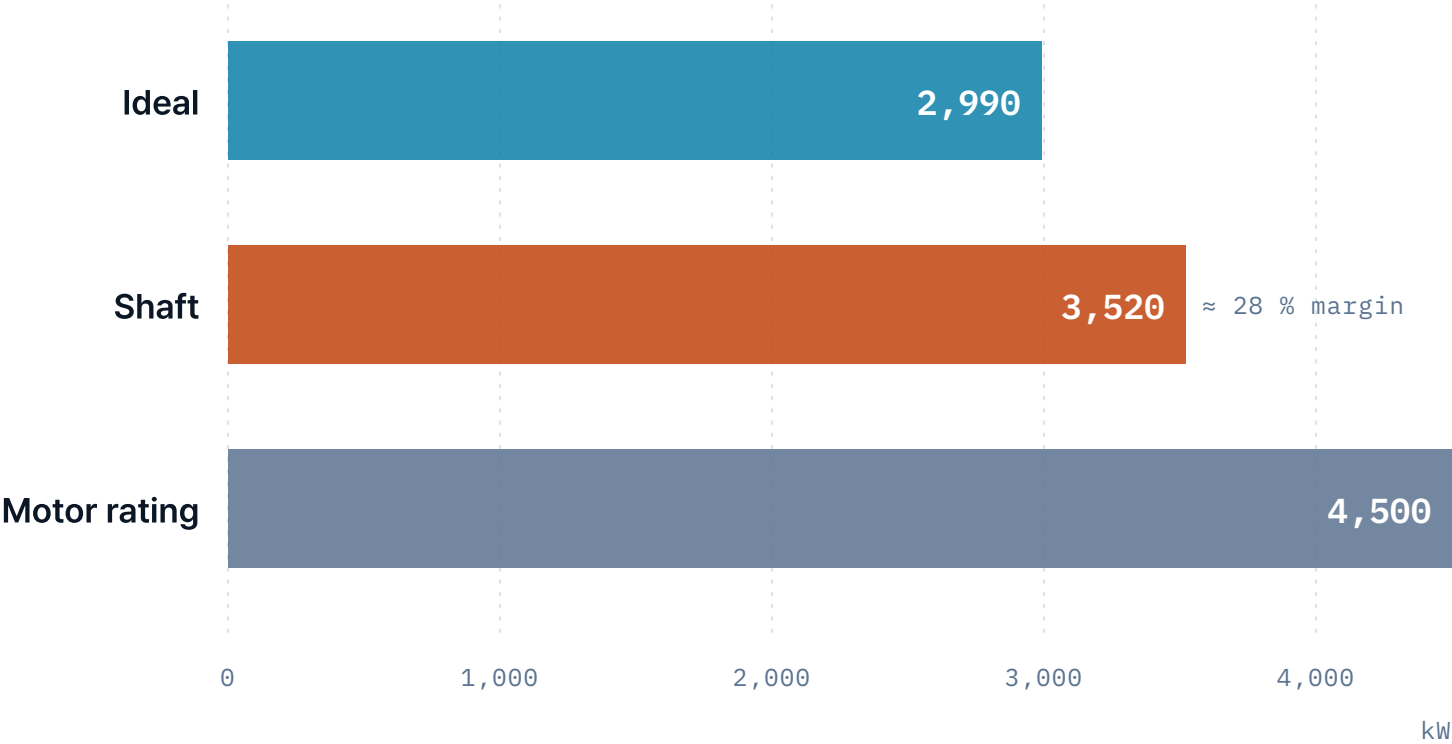
7.0 MPa

volumetric flow $\dot{V}$ the volume pushed per second = mass flow/density · **circulator (blower)** the rotating machine that pushes a gas, the gas-side counterpart of a pump

Calculation This textbook §2.2 · $\dot{m} = Q/(c_p \Delta T) = 250 \text{ MW}/(5195 \times 500) = 96.2 \text{ kg/s}$ · inlet 250 °C · $\rho = P/(RT)$ · friction loss $\propto \dot{m}^2/\rho$, blower power $\propto \dot{m}^3/\rho^2$ · multiples relative to 7 MPa

Checking the published HTR-PM circulator specifications

- Thirty lines of arithmetic show the four published figures **do not contradict each other** - chew on every dataset like this once
- 3.5 MW of shaft power is **1.4 %** of the thermal output and all of it ends up in the cold helium. Leave it out and **the core inlet comes out about 7 K too low**



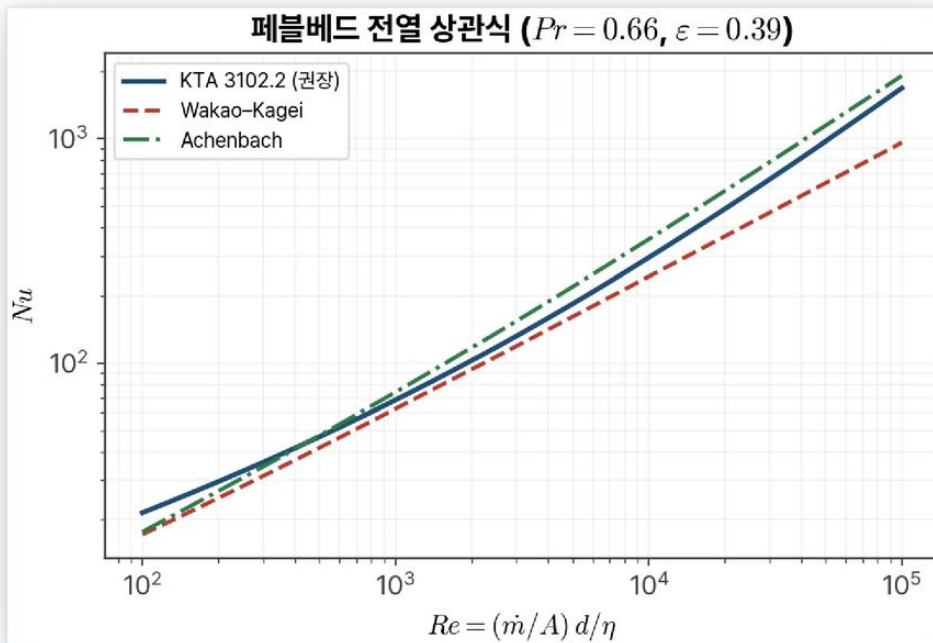
QUANTITY	VALUE
Mass flow $\dot{m}$	96.2 kg/s
Density ρ (7 MPa, 250 °C)	6.44 kg/m³
Volume flow $\dot{V}$	14.9 m³/s
Ideal power $\dot{V}\Delta p$	2 990 kW
Shaft power ($\eta = 0.85$)	3 520 kW
Motor rating	4 500 kW

shaft power the power the shaft must actually deliver = ideal power/efficiency · **adiabatic efficiency** ratio of ideal to actual compression work

Source This textbook §2.3 · published specs: 200 kPa rise, 250 °C suction, 4000 rpm, 4500 kW motor · adiabatic efficiency 0.85 is assumed · incompressible approximation

Pebble-bed correlations and the prismatic correction

- Flow threads between spheres, so pipe correlations do not apply - **KTA 3102.2** is the de facto standard ($100 < Re < 10^5$, $0.36 < \varepsilon < 0.42$, $\pm 20\%$)
- Prismatic channels are pipes, but heavy heating changes wall properties - raw Dittus–Boelter **overpredicts by 18-32 %**; the Taylor correction is standard



Pebble-bed correlations compared ($Pr = 0.66$, $\varepsilon = 0.39$) - KTA fits data at low Re , Wakao-Kagei at high Re · textbook Fig. 2.2

KTA 3102.2

$$Nu = 1.27 Pr^{1/3} \varepsilon^{-1.18} Re^{0.36} + 0.033 Pr^{1/2} \varepsilon^{-1.07} Re^{0.86}$$

Prismatic · Taylor

$$Nu_b = 0.023 Re_b^{0.8} Pr_b^{0.4} \times (T_s/T_b)^{-0.57-1.59D/x}$$

LAMINAR FLOOR - ALWAYS KEEP IT

$$Nu = 4.364$$

When forced flow stops Re falls to a few hundred; a turbulent correlation alone sends Nu to zero and the calculation overheats unrealistically

KTA the German nuclear safety standards commission · **porosity** ε the fraction of bed volume between pebbles · **Nu** Nusselt number, dimensionless convective heat transfer · **Re** Reynolds number

Source This textbook §2.4, Fig. 2.2 · KTA 3102.2 (heat transfer), 3102.3 (pressure drop) · Taylor fits 98 % of helium and nitrogen data within $\pm 25\%$

Irradiated graphite - size, conductivity and Wigner energy

- Graphite is **a consumable** - it shrinks, then past the turnaround it swells. That turnaround sets the component life
- Conductivity falls **86 %** by 25 dpa, most of it **before 1 dpa**. Solving conduction cooling with unirradiated values badly inflates the margin

EFFECT	VALUE (IG-110)	WHAT IT MEANS FOR ANALYSIS
Dimensional turnaround	600 °C · ~15 dpa · -7 % volume	sets component life
Thermal conductivity	130-160 → -86 % at 25 dpa	top uncertainty in peak accident fuel temperature (MELCOR)
Wigner energy · irradiated at 30 °C	up to 630 cal/g	a low-temperature problem
Wigner energy · irradiated at 450 °C	15 cal/g	not an issue at HTGR temperatures (250-1100 °C) - an inference

CONDUCTIVITY LOSS · 25 DPA

-86 %

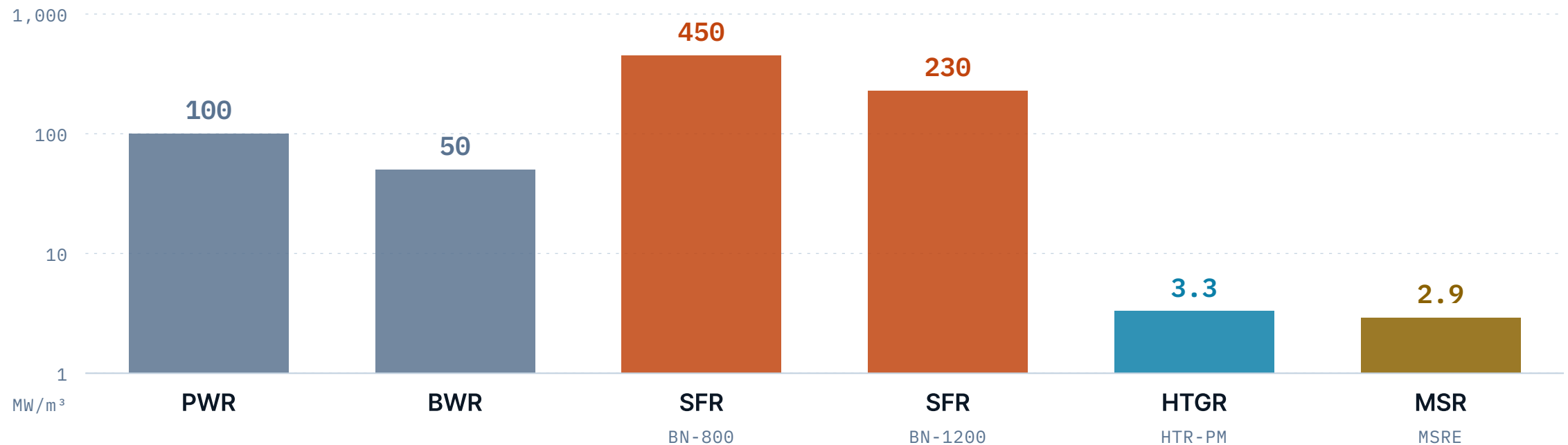
That Wigner energy is harmless in an HTGR is **a well-grounded inference, not a measurement**. ORNL says outright that no release-rate data exist for graphite irradiated at HTGR temperatures.

dpa displacements per atom, a measure of irradiation dose · **turnaround** the dose where shrinkage turns to swelling · **Wigner energy** energy stored in the lattice by irradiation that can be released at once · **MELCOR** the US severe-accident code

Source This textbook §2.5 · IG-110 values · ORNL states there is no measured release-rate data for graphite irradiated at HTGR temperatures

Average core power density by reactor type

- HTR-PM's 3.3 MW/m³ is **about 1/30** of an LWR - **by intent**, not failure. Spreading decay heat over a large graphite mass lets conduction and radiation alone remove it
- The price is volume - **78 m³** of core for 250 MWt against **2.5 m³** for an LWR. **Passive safety is not free; it is bought with volume**



power density thermal power per unit core volume · BN-800, BN-1200 Russian sodium-cooled fast reactors · MSRE the 1960s ORNL Molten Salt Reactor Experiment

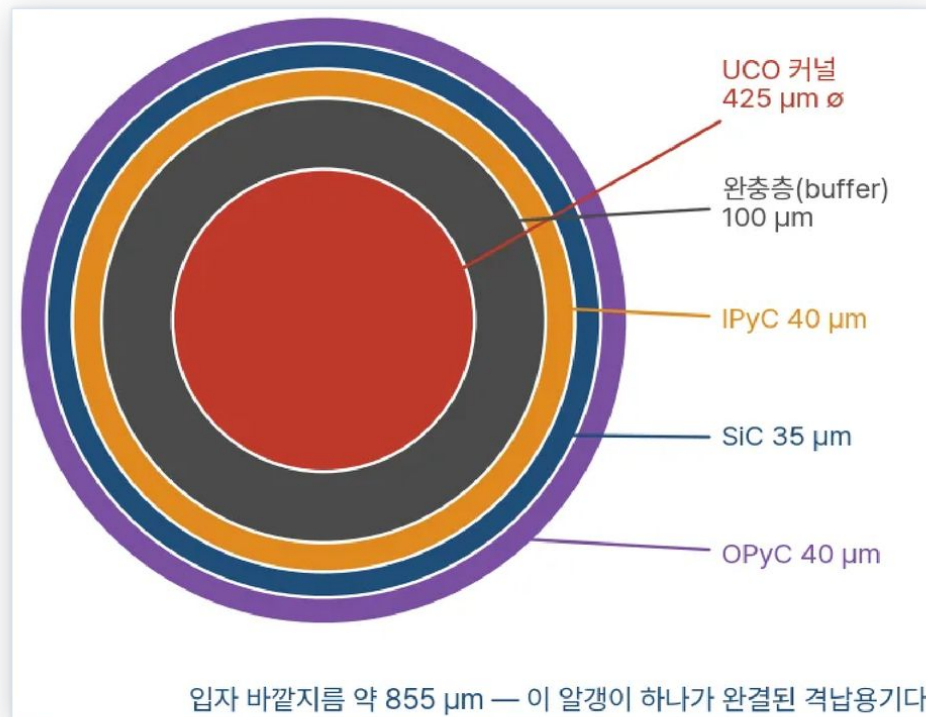
Source This textbook §2.6, Fig. 2.3 · MW/m³ · log scale · HTR-PM core $\varnothing$ 3 m × 11 m = 78 m³

03 TRISO

A light-water reactor puts the fission-product barrier in the cladding and the building. The HTGR moves it **inside the fuel grain**. The AGR tests show that this rests on data, not on instinct.

TRISO layers, and where containment moves to

- An LWR has four barriers: pellet, cladding, pressure boundary, containment. The HTGR moves them **inside a particle under 0.9 mm across** - about **5 billion** per module
- The release limit is set by **particle statistics**, not building integrity → a low-pressure confinement becomes possible



LAYER	THICKNESS	ROLE
Kernel	$\varnothing$ 425~500 μm	where fission happens
Buffer	~100 μm	holds fission gas, absorbs swelling
IPyC	40 μm	keeps kernel species off the SiC
SiC	35 μm	the real pressure boundary, stops metals
OPyC	40 μm	protects SiC, bonds to matrix

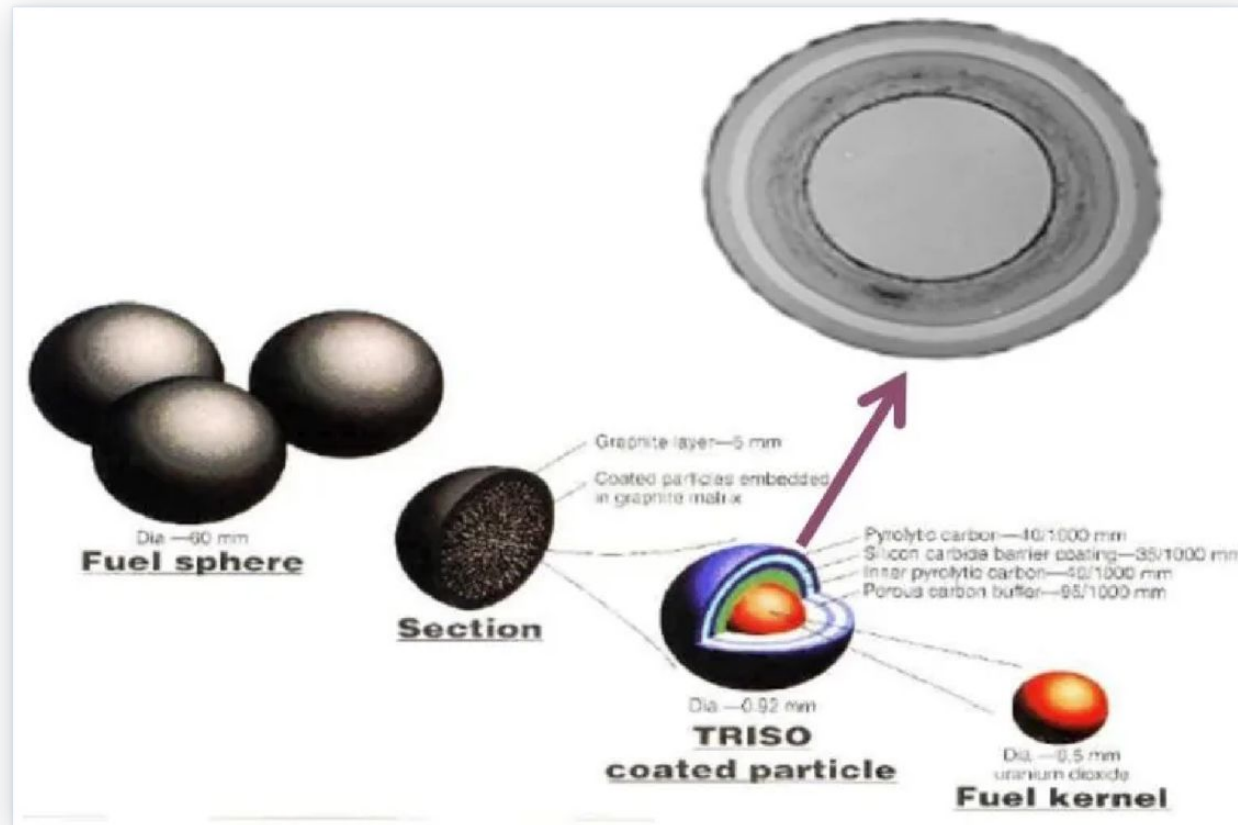
A TRISO particle in section (AGR-2 UCO) - each grain a complete containment · textbook Fig. 3.1

UCO uranium oxycarbide kernel · PyC pyrolytic carbon · IPyC, OPyC inner and outer dense pyrocarbon · SiC silicon carbide · **confinement** a low-pressure building in place of a pressure-retaining containment

Source This textbook §3.1, Fig. 3.1, Table 3.1 · AGR-2 UCO dimensions · particle outer diameter about 855 μm

From pebble to TRISO particle - the chain of scales

- Each 60 mm graphite pebble holds **about 12,000** TRISO particles (HTR-PM) embedded in a graphite matrix
- The outer 5 mm is fuel-free graphite - so particles are not exposed when pebbles knock together



PEBBLE · FUELLED ZONE

60 mm · 50 mm

PARTICLES PER PEBBLE (HTR-PM)

~12 000

HEAVY METAL PER PEBBLE

7.0 g

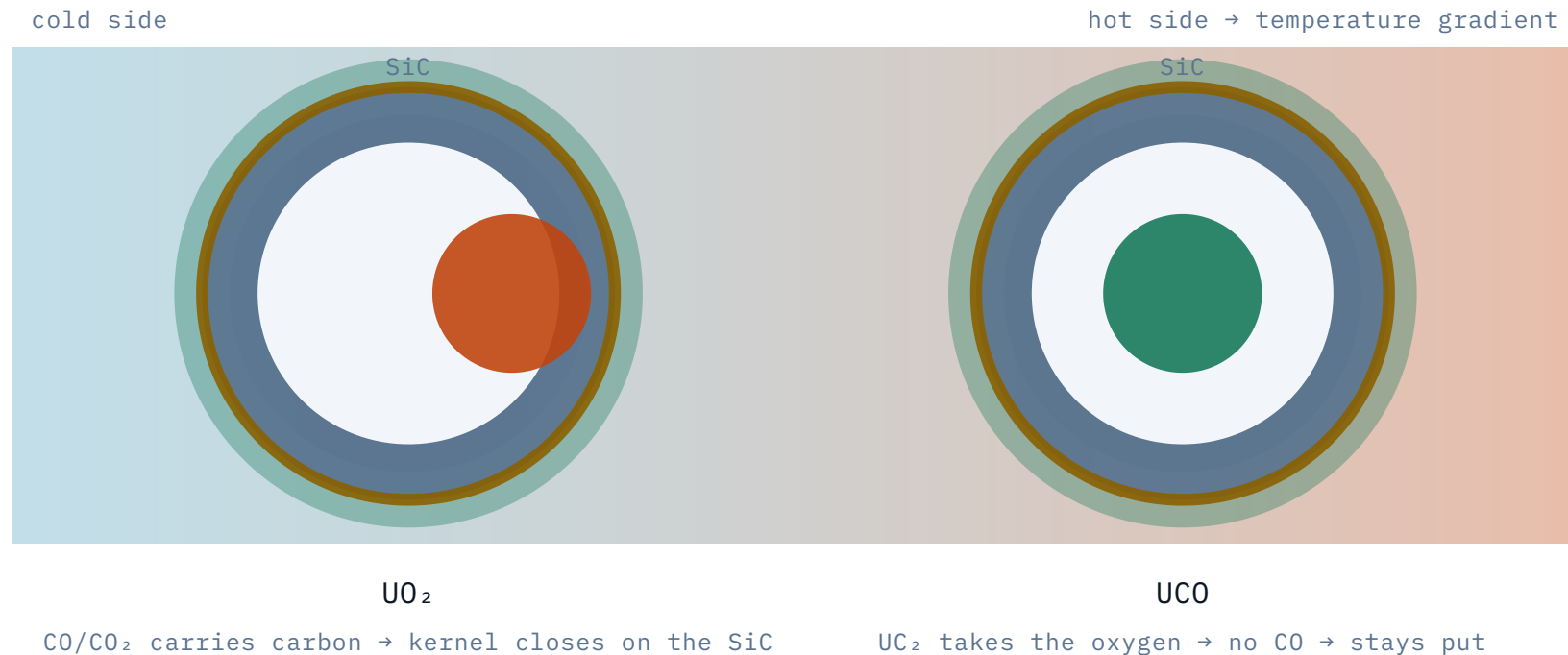
PEBBLES PER MODULE (EQUILIBRIUM)

420 000

Fuel sphere (ø 60 mm) → section → TRISO coated particle (ø ~0.92 mm) → fuel kernel (ø ~0.5 mm) · textbook Fig. 3.2

SiC stress and the CO amoeba effect - why UCO

- A thicker buffer gives the gas more room and lowers p ; a thicker SiC lowers σ_θ - **high burnup needs both**
- In practice CO comes first - it raises pressure and walks the kernel toward the hot side. **In UCO the carbide (UC_2) takes up the excess oxygen, so CO never forms**



$$\sigma_\theta = p r / (2 t_{\text{SiC}})$$



TWO HARMS FROM CO

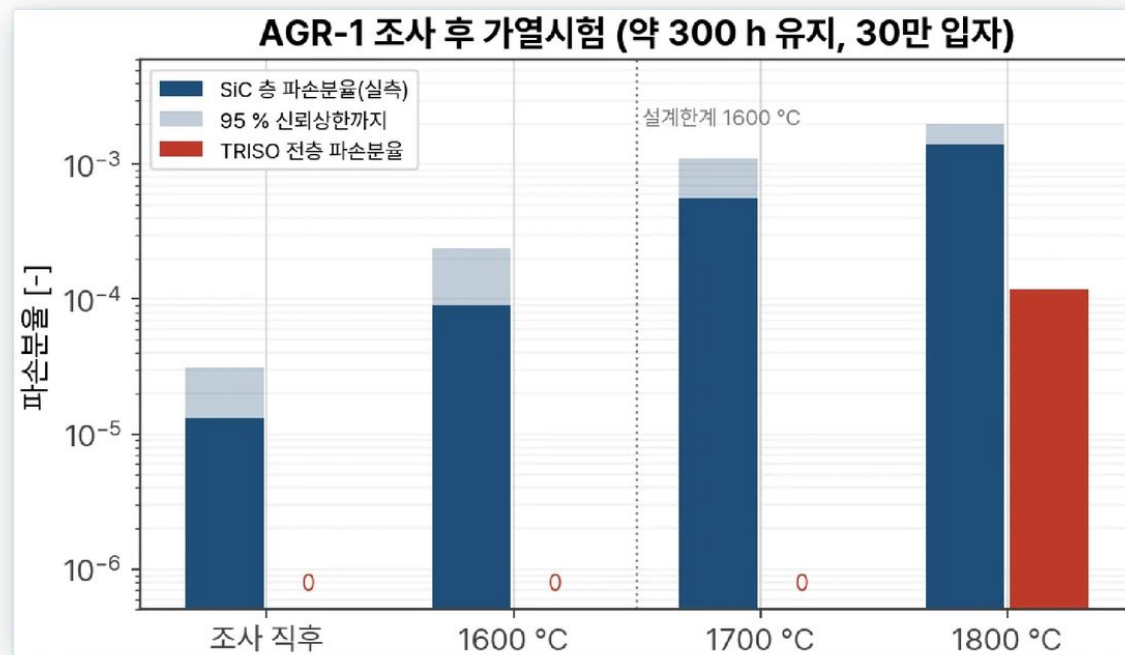
- ① higher pressure
- ② kernel migration → SiC contact

hoop stress σ_θ the circumferential tensile stress in a spherical shell · **amoeba effect** CO/CO₂ carrying carbon along the temperature gradient so the kernel migrates toward the hot side · **UC₂** uranium dicarbide

Source This textbook §3.2 · thin spherical-shell approximation · the drawing is a concept sketch of the amoeba effect (sizes and speed not to scale)

AGR-1 post-irradiation heating - SiC failure fraction against temperature

- Read the **slope**, not the numbers - 1600 → 1800 °C, 200 K, and the failure fraction rises **15-fold**. The 1600 °C limit sits just before the curve bends
- No** TRISO (three-layer) failures in irradiation. Even at 1800 °C, only 2 of 16,500 failed through all layers



CONDITION	SIC FAILED	95 % BOUND	ALL
As irradiated	1.3×10^{-5}	3.1×10^{-5}	0
1600 °C	9.1×10^{-5}	2.4×10^{-4}	0
1700 °C	5.6×10^{-4}	1.1×10^{-3}	0
1800 °C	1.4×10^{-3}	2.0×10^{-3}	2/16 500

AGR-1 heating tests (about 300 h, 300,000 particles) - blue SiC failure (measured), grey 95 % upper bound, red all-layer failure · textbook Fig. 3.3

AGR the US Advanced Gas Reactor fuel programme · ATR INL's Advanced Test Reactor · EFPD effective full-power days · FIMA fissions per initial metal atom · FACS the post-irradiation heating furnace

Source This textbook §3.3, Table 3.2, Fig. 3.3 · AGR-1 (2006-2009, 620 EFPD) · about 300,000 UCO particles · 11.3-19.6 % FIMA · time-averaged peak 1069-1197 °C · 300 h in the FACS furnace

AGR-2 and AGR-5/6/7, and TRISO's weak spot - silver (Ag-110m)

- Never quote the number alone - the 180-440 failures in AGR-5/6/7 capsule 1 were **a gas-line leak in the rig**, not fuel performance
- Say "TRISO is perfect" in a talk and the question about **silver** is certain to come

AGR-2

UCO and UO₂ in one capsule

irradiated and heated alike

UCO - SiC fails only at 1800 °C

UO₂ - several already at 1600 °C

clear CO attack on the SiC

→ the chemistry is confirmed

AGR-5/6/7

licensing qualification (2018-2020)

about 570,000 particles

peak burnup 15.3 % FIMA

peak time-averaged 1432 °C

capsules 4 and 5 - zero failures

capsule 1, 180-440 = rig defect

Silver Ag-110m

the one species SiC cannot hold

AGR-1 capsule release 0.012-0.38

about 7 % in a 1695 °C transient

irrelevant to safety limits

deposits on heat-exchanger surfaces

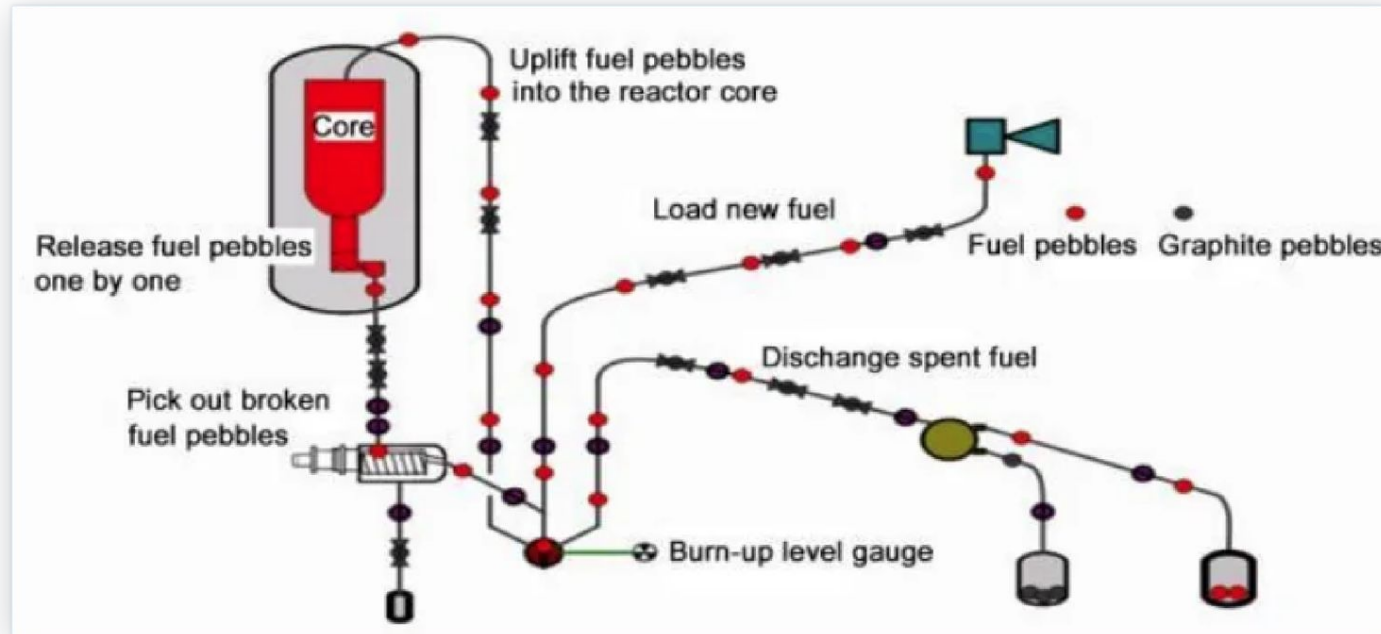
→ drives maintenance dose

Ag-110m a radioactive silver isotope that passes through SiC · **qualification test** proving performance under set conditions for licensing · **maintenance dose** radiation dose to maintenance workers

Source This textbook §3.3 · the capsule 1 failures in AGR-5/6/7 (2018-2020) were reported as a gas-line leak in the test rig

Pebble recirculation and online refuelling - three gains, three costs

- **Gains** - fresh fuel every day keeps excess reactivity small, no shutdowns, flat burnup
- **Costs** - complex machinery, graphite dust from broken pebbles, and fuel positions known **only statistically**



Pebble circuit - discharge → broken-pebble sorting → burnup gauge → reload or discard · Fig. 3.4



Full-scale fuel-handling test rig · Fig. 3.5

online refuelling loading and unloading fuel without shutting down · **excess reactivity** reactivity beyond what criticality needs · **prismatic** an HTGR with fuel in hexagonal graphite blocks · **Gwd/tU** thermal energy per tonne of uranium (burnup)

Source This textbook §3.4, Figs. 3.4-3.5 · HTR-PM pebbles pass the core about 15 times on average and leave at 90 GWd/tU

Three regimes of **graphite oxidation**, by temperature

- Graphite reacts with both air and steam. The controlling mechanism is entirely different in each temperature range
- Using **one Arrhenius law across all ranges without first deciding the regime gives badly wrong answers**

about 600-750 °C

Regime 1 · chemical control

oxygen penetrates 3-4 mm

attacks the interior evenly

high activation energy, T-sensitive

NBG-18 187 · IG-110 222 kJ/mol

intermediate

Regime 2 · pore diffusion

diffusion in pores sets the rate

activation energy halves

shallower penetration

above 900 °C

Regime 3 · boundary layer

only the surface erodes

almost no T-dependence

oxygen supply is the limit

activation energy sets how sensitive a reaction rate is to temperature · **Arrhenius law** $\text{rate} = A \cdot \exp(-E/RT)$ · **boundary-layer mass transfer**
oxygen supply through the gas layer next to the surface

Source This textbook §3.5 · activation energy 187 kJ/mol for NBG-18 and 222 kJ/mol for IG-110 (regime 1)

The TRISO fuel supply chain - China, the US and Korea

- TRISO is a **supply-chain problem**, not a technology problem. China is past it, the US is catching up, and **Korea has no commercial production**

Commercial fuel plant, 300,000/a, Baotou, CNNC fuel plant

- 2013/03/ started construction
- 2016/03/ finished plant installation and commission
- 2016/08/ started production
- 2017/12/ 300,000 fuel pebbles produced



38

The Baotou commercial pebble plant - construction 2013, 300,000 pebbles by December 2017 · textbook Fig. 3.6

CHINA · BAOTOU

300,000 pebbles a year; 300,000 made by Dec 2017

US · TRISO-X TX-1 (OAK RIDGE)

700,000 pebbles, 5 t U a year - eleven Xe-100s. Shell done Aug 2026, fit-out under way

KOREA

No commercial production. Whether the ₩44.5 bn public-private project (from July 2024) fills the gap is the question

HALEU uranium enriched to 5-20 % · TRISO-X X-energy's fuel subsidiary · **Category 2** the US facility class for moderate-strategic-significance material

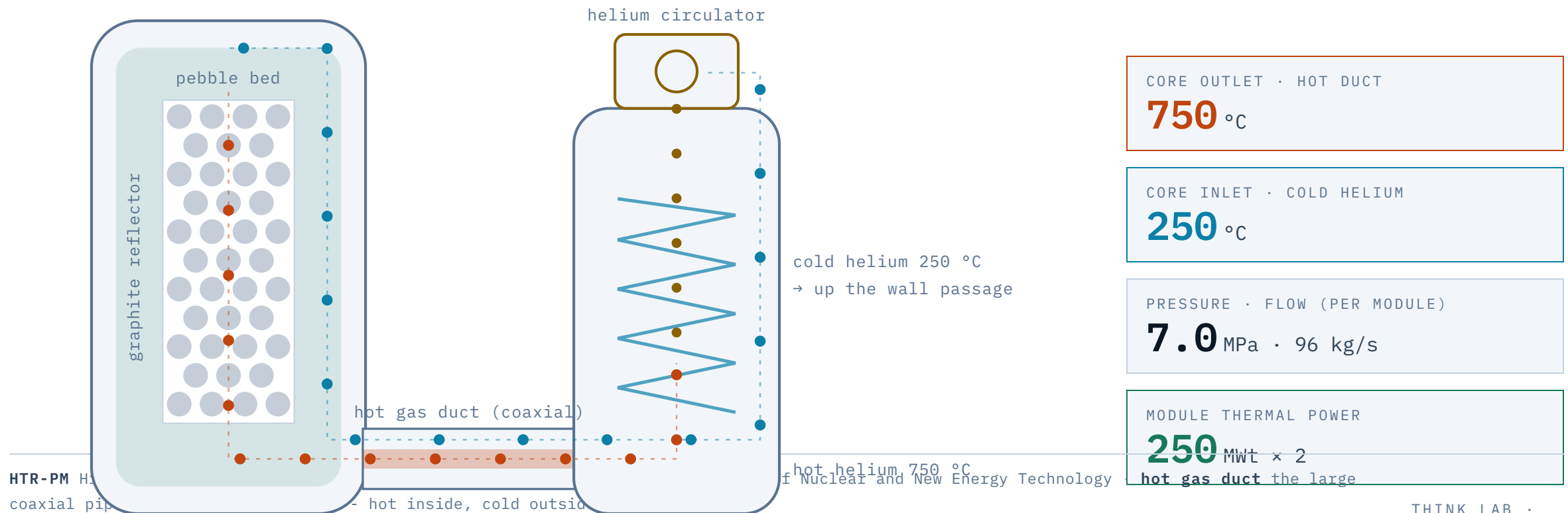
Source This textbook §3.6, Fig. 3.6 · as of August-September 2026 · TX-1 holds the first US Category 2 licence for a HALEU fuel plant

04 Taking HTR-PM apart

The only Generation IV reactor in commercial operation, component by component.
We check by hand that the published design values agree, and separate the brochure's "rated" from how it actually runs.

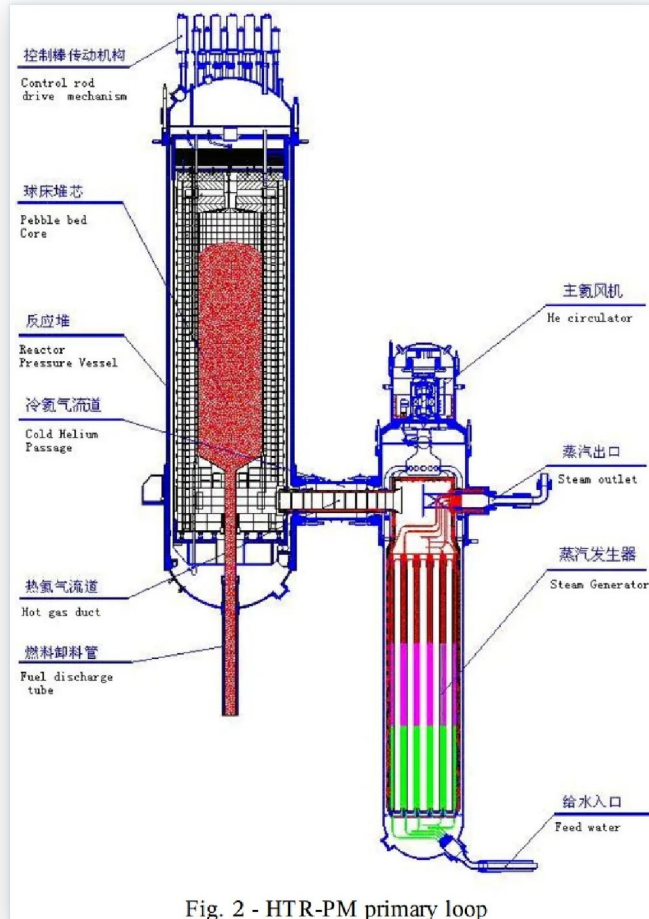
The helium flow path in one HTR-PM module

- Two 250 MWt modules share one turbine - 500 MWt / 211 MWe gross. **Grid December 2021, commercial 5 December 2023** - the first commercial Gen IV plant
- Cold helium **rises along the vessel wall**, turns at the top, and **heats up flowing down** through the pebbles to the hot gas duct at the bottom



Source This textbook §4.1 The drawing is a concept sketch of the flow order, not to scale (Shidao wan, Shandong · designed by Tsinghua INET · descended from HTR-10 (10 MWt, critical December 2000))

Three features of the primary-loop layout



The HTR-PM primary loop - textbook Fig. 4.1

1 Vessel and SG stand side by side

Unlike an LWR the SG is not above the core; a large duct joins them. A break in that duct is this reactor's LOCA.

2 Cold helium washes the vessel wall first

The vessel stays near 250 °C around a 750 °C core, which is why ordinary vessel steel (SA-508 family) will do.

3 Helium flows top to bottom

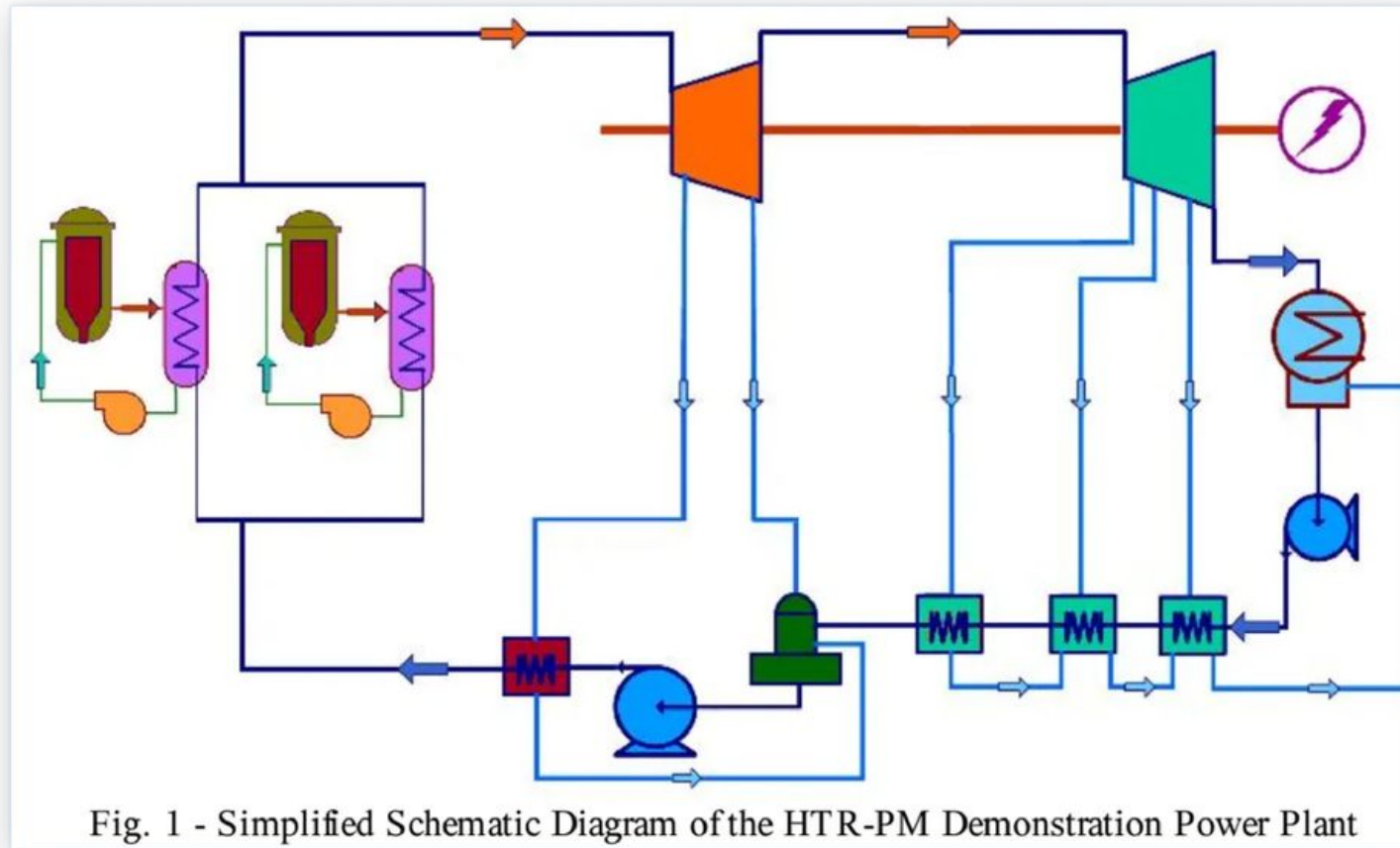
The pebble bed is pressed down and stable. When forced flow stops, buoyancy acts upward, against it, and flow halts by itself.

SA-508 the low-alloy steel common in reactor pressure vessels · **LOCA** loss-of-coolant accident · **buoyancy** the lift that density differences produce

Source This textbook §4.1, Fig. 4.1 · reactor vessel on the left, steam generator on the right, circulator on top of the steam generator

Plant schematic - two reactors share **one turbine**

- Each module has its own reactor, steam generator and circulator. Steam from both goes to **one turbine**
- Steam at **13.25 MPa / 567 °C** - far better than an LWR's saturated steam near 7 MPa, hence the higher efficiency



Simplified HTR-PM demonstration plant - two modules on the left, turbine-generator in the middle, feedwater heaters on the right · textbook Fig. 4.2

HTR-PM principal design data

ITEM	VALUE	NOTE
Module thermal power	250 MWt × 2	
Electrical output	211 MWe gross / 150 MWe net	one shared turbine
Helium temperature · pressure	250 → 750 °C · 7.0 MPa	6.8 MPa in the 2023 tests
Helium flow	96 kg/s per module	checked with $\dot{m} = Q/(c_p \Delta T)$
Core size	∅ 3.0 m × 11.0 m, 77.8 m³	
Power density	3.3 MW/m³	about 1/30 of an LWR
Pebbles	∅ 60 mm · 420,000 per module	50 mm fuelled zone
Enrichment · burnup	8.5 % · 90 GWd/tU	7.0 g heavy metal per pebble
Steam conditions	13.25 MPa / 567 °C	14.1 MPa / 570 °C in some sources
Temperature coefficient	about -5×10^{-5} /K	negative
Fuel temperature limit	1600 normal · 1620 °C accident	about 1200 °C peak in normal operation

enrichment the U-235 share of the uranium · **temperature coefficient** reactivity change per kelvin - negative means self-limiting · **MWe, MWt** electrical and thermal megawatts

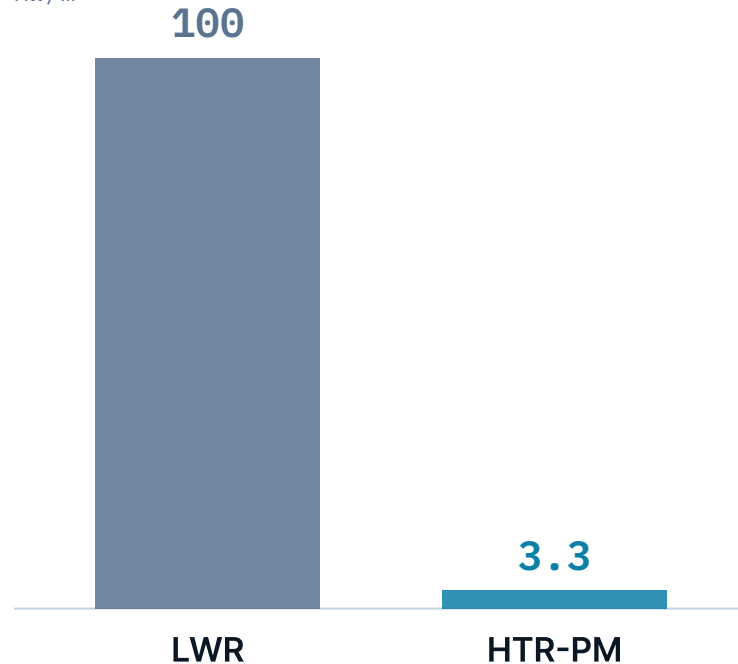
Source This textbook §4.2, Table 4.1 · steam is also quoted as 14.1 MPa / 570 °C at the turbine inlet · the 2023 tests ran at 6.8 MPa

Power density and **specific power** - per volume and per mass

- Per volume it is 1/30 of an LWR, yet per mass of uranium it is **more than twice as high** - because graphite fills most of the core
- Mix the two and the conclusion goes astray. The 2.9-year residence also matches the "about 3 years" in presentations

Per volume · power density

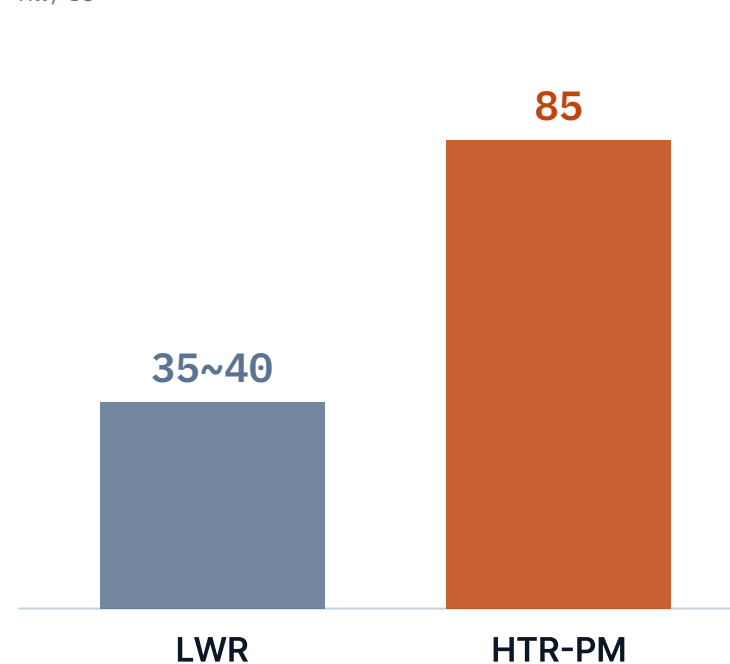
MW/m³



about 1/30

Per mass · specific power

MW/tU



more than 2x

$$420\,000 \times 7.0 \text{ g} = 2.94 \text{ tU}$$

$$250 \text{ MW} / 2.94 \text{ tU} = 85 \text{ MW/tU}$$

$$90\,000 / 85 = 1\,059 \text{ d}$$

CORE RESIDENCE

2.9 years

ONE PASS (15 ON AVERAGE)

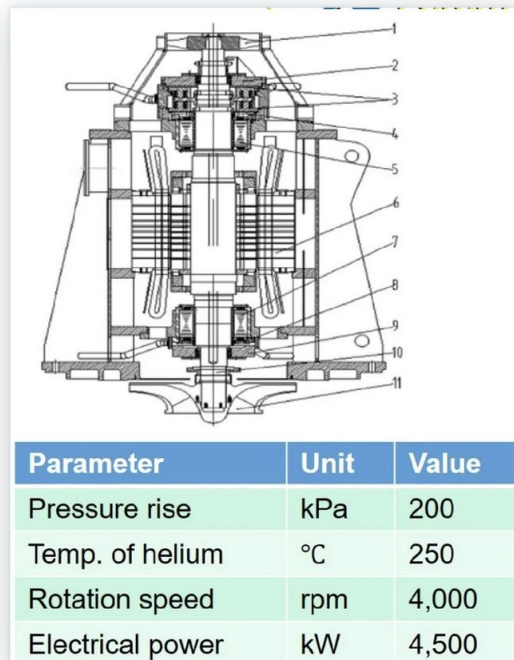
~70 days

power density thermal power per core volume (MW/m³) · **specific power** thermal power per fuel mass (MW/tU) · **residence time** total time the fuel spends in the core

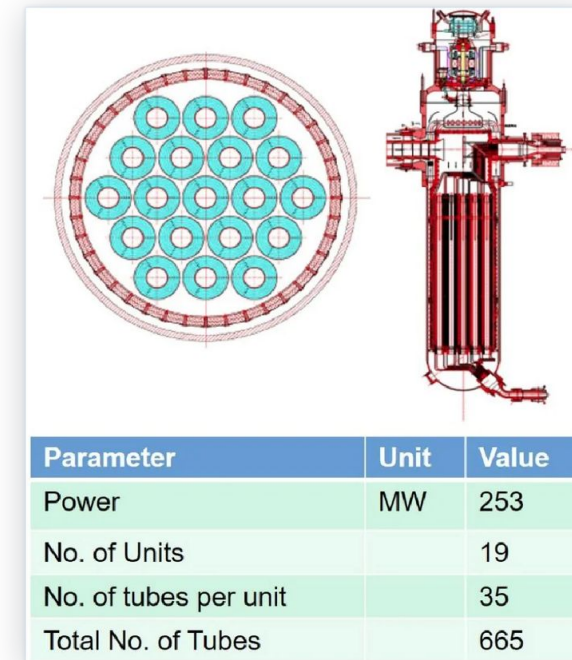
Calculation This textbook §4.2 · $420,000 \times 7.0 \text{ g} = 2.94 \text{ tU}$ · $250 \text{ MW} / 2.94 \text{ tU} = 85 \text{ MW/tU}$ · $90,000 \text{ MWd/tU} \div 85 \text{ MW/tU} = 1,059 \text{ d} \approx 2.9 \text{ years}$ · 15 passes on average → about 70 days a pass

The helium circulator and the steam generator

- The circulator shaft floats on **active magnetic bearings** - no oil, so no route for oil into the primary circuit. In a gas reactor oil vapour means blockage
- The steam generator sits **inside the primary boundary** - a tube break lets water in, and $C + H_2O \rightarrow CO + H_2$. The 19 units exist for isolation (ch. 5)



Circulator section and data · Fig. 4.3



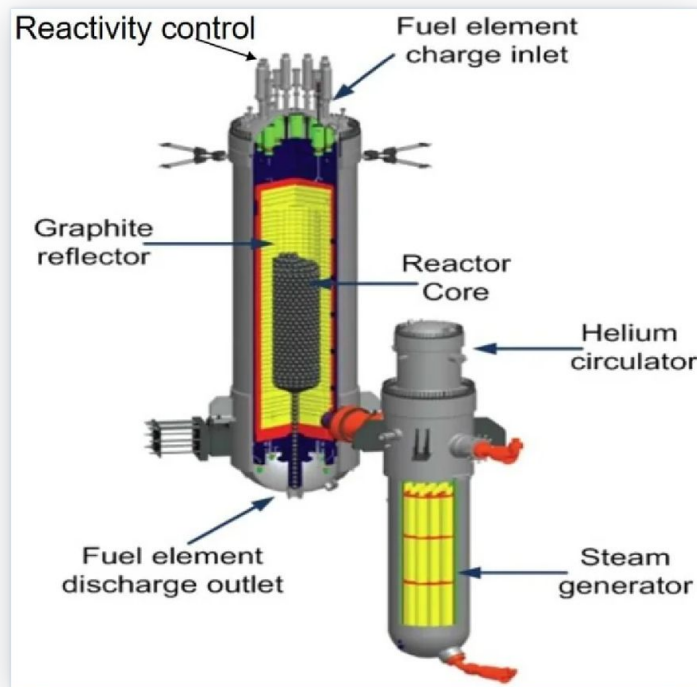
Steam generator - 19 units, 665 tubes · Fig. 4.4

AMB active magnetic bearing - electromagnets float the shaft, no lubricating oil · **helical coil** heat-transfer tubes wound in a spiral · **water ingress** water entering the primary circuit through a failed tube

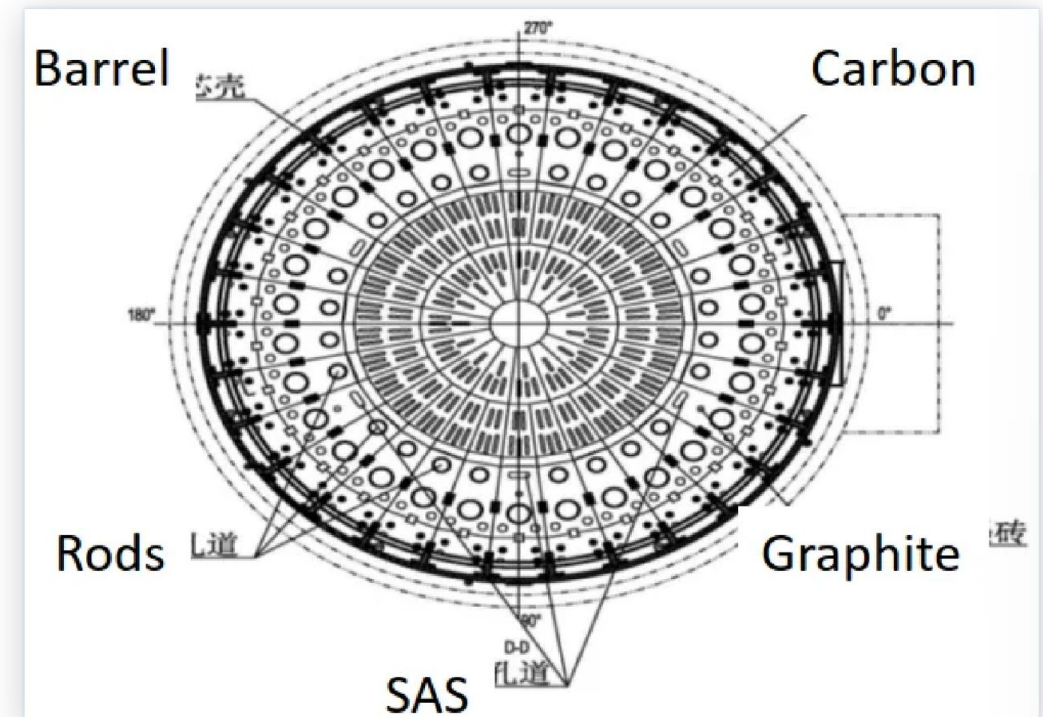
Source This textbook §4.3, Figs. 4.3-4.4 · circulator 200 kPa, 250 °C, 4000 rpm, 4500 kW · steam generator 253 MW, 19 units × 35 tubes = 665 tubes

The vessel and core internals - rods and SAS in the reflector

- The core is a pile of pebbles, so nothing can be inserted into it - rods and small absorber spheres **both go into holes in the graphite reflector**
- Rods for operation and shutdown; **SAS for shutdown margin over long outages** - two independent means of reactivity control



Vessel cutaway - reactivity control, fuel inlet, reflector, core, circulator, steam generator, discharge · Fig. 4.5



Core cross-section - barrel, carbon, graphite, rods, SAS · Fig. 4.6

SAS small absorber sphere system - absorber balls dropped into reflector holes · **shutdown margin** negative reactivity to spare when shut down · **core barrel** the metal cylinder round the core structure

Source This textbook §4.3, Figs. 4.5-4.6 · the core section runs outward from barrel to carbon blocks, graphite reflector, rod holes and SAS holes

The top of the core, and the vessel in transit

- Above the pebble bed the **rod guide tubes** in the reflector are visible - at the edge, not inside the core
- Because the core is held to 3 m across, the vessel is slim enough → **to travel by road**. Why the diameter is capped comes in ch. 5



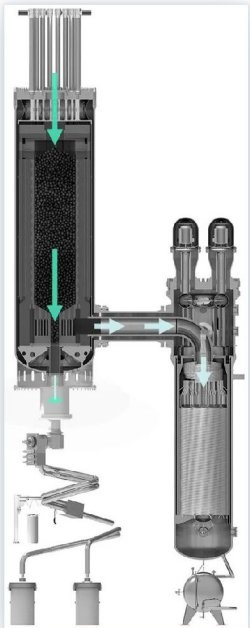
Top of the core - pebbles and rod tubes · Fig. 4.7



The reactor vessel in transit · Fig. 4.8

Xe-100 (X-energy) - design data and licensing

- Furthest along in US licensing - and its first customer is a **Dow chemical plant** (Seadrift, Texas; Long Mott), not a utility. The industry expects a permit in early 2027
- **Core diameter, height, power density, inlet temperature and flow are withheld in the NRC filings** - if a slide lists them, someone almost certainly guessed



Xe-100 cutaway · Fig. 4.9

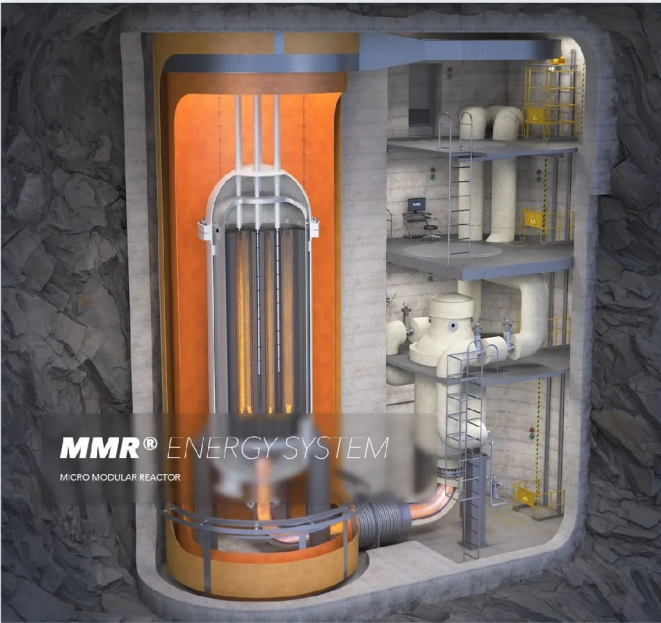
ITEM	XE-100	HTR-PM
Module output	200 MWt / 80 MWe	250 MWt
Standard plant	4 modules · 320 MWe	2 modules · 211 MWe
Helium pressure	6 MPa	7 MPa
Outlet · steam	750 °C · 565 °C / 16.5 MPa	750 °C · 567 °C / 13.25 MPa
Pebbles · kernel	~220,000 · UCO 425 µm	420,000 · UO ₂
Enrichment · discharge burnup	15.5 % · 163 Gwd/MTU	8.5 % · 90 Gwd/tU
Load following	40 → 100 % in 12 min	—

FONSI Finding of No Significant Impact - the environmental assessment outcome · **MTU** metric tonne of uranium · **load following** raising and lowering output with demand

Source This textbook §4.4, Fig. 4.9 · **as of September 2026** · Long Mott construction permit filed March 2025, docketed May 2025 · FONSI May 2026 · final safety evaluation due November 2026

Other HTGRs - KRONOS MMR, BANR, HTTR, Korea and the UK

- **“MMR = 15 MWt” is the old USNC design**; the KRONOS docketed at NRC is 45 MWth - both true, but cite which. This mix-up is very common in non-LWR material



MMR below-grade layout concept · Fig. 4.10

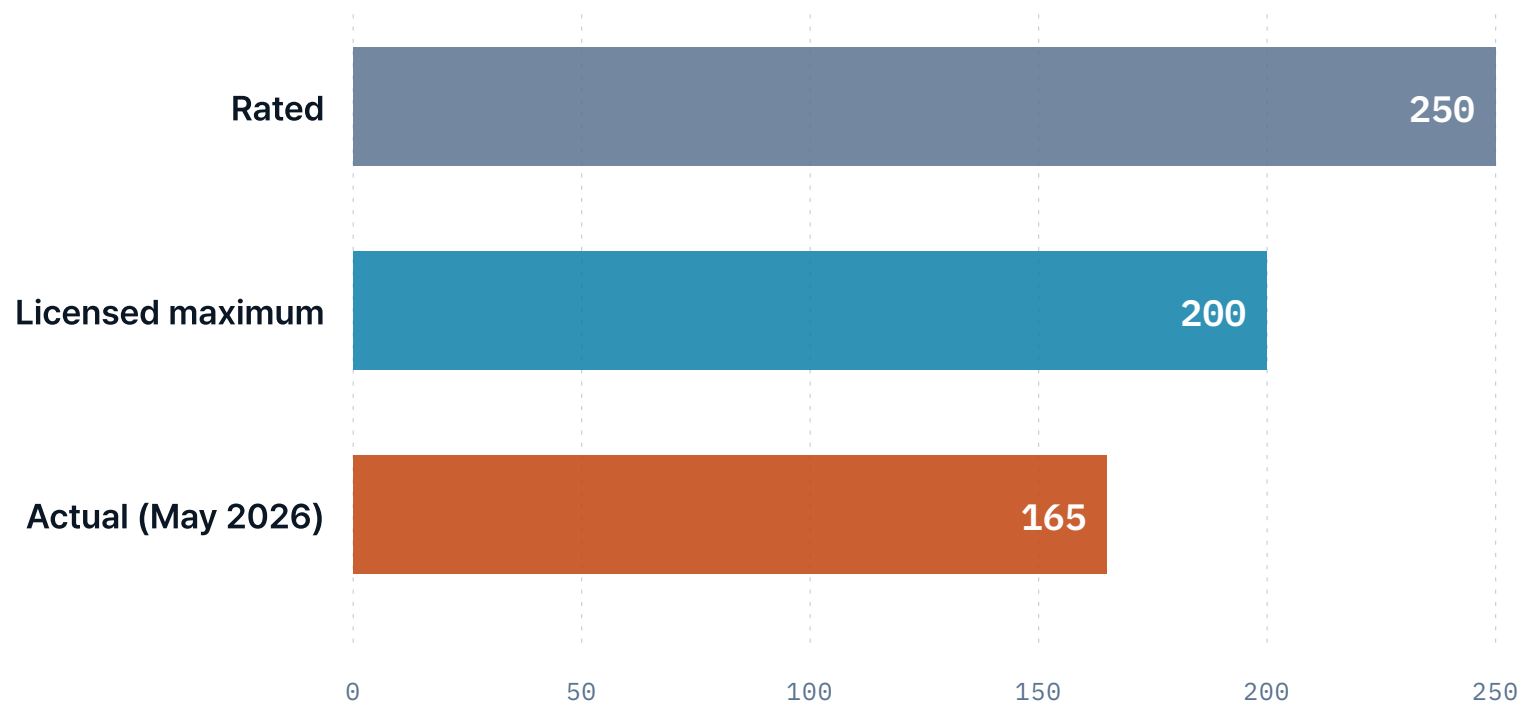
DESIGN	SIZE	STATUS (SEPT 2026)
KRONOS MMR	45 MWth (as docketed)	UIUC filed a construction permit March 2026, docketed May. Helium primary with molten-salt storage, FCM fuel
BWXT BANR	75 MWt / 20 MWe	Picked for the US Army Janus programme in Aug 2026, Fort Campbell. Construction target late 2028. HALEU TRISO, 4-year refuelling, road or rail
HTTR (Japan)	30 MWt test reactor	Prismatic, critical 1998. World-first 950 °C outlet at full power, April 2004. What matters is the testing in ch. 5
Korea	750-950 °C target	2004 NHDD never reached a demonstrator. ₩44.5 bn public-private project from July 2024, to 2027. KEPCO E&C-POSCO E&C MOU in Sept 2026
UK	AMR demonstrator	Chose the HTGR in 2021-22. Two MOUs with JAEA, NNL and Rolls-Royce in June 2026. No site, vendor or build decision

MMR Micro Modular Reactor · FCM Fully Ceramic Microencapsulated fuel · BANR BWXT Advanced Nuclear Reactor · NHDD Korea’s nuclear hydrogen demonstration · AMR the UK Advanced Modular Reactor programme · MOU memorandum of understanding

Source This textbook §4.4, Fig. 4.10 · as of September 2026 · USNC filed Chapter 11 in October 2024; NANO Nuclear bought the IP for \$8.5 M

HTR-PM **rated output** against actual operation

- Both modules run at **165 MWt**, about 116 MWe to the grid. The present licence limit is 200 MWt, with uprating planned once the core reaches equilibrium in 2-3 years
- This is normal for a first-of-a-kind, not a failure. But **"Gen IV is commercial"** and **"running steadily at rated power"** are very different sentences



CUMULATIVE CAPACITY FACTOR (WNA)

29.7 %

GRID OUTPUT (MAY 2026)

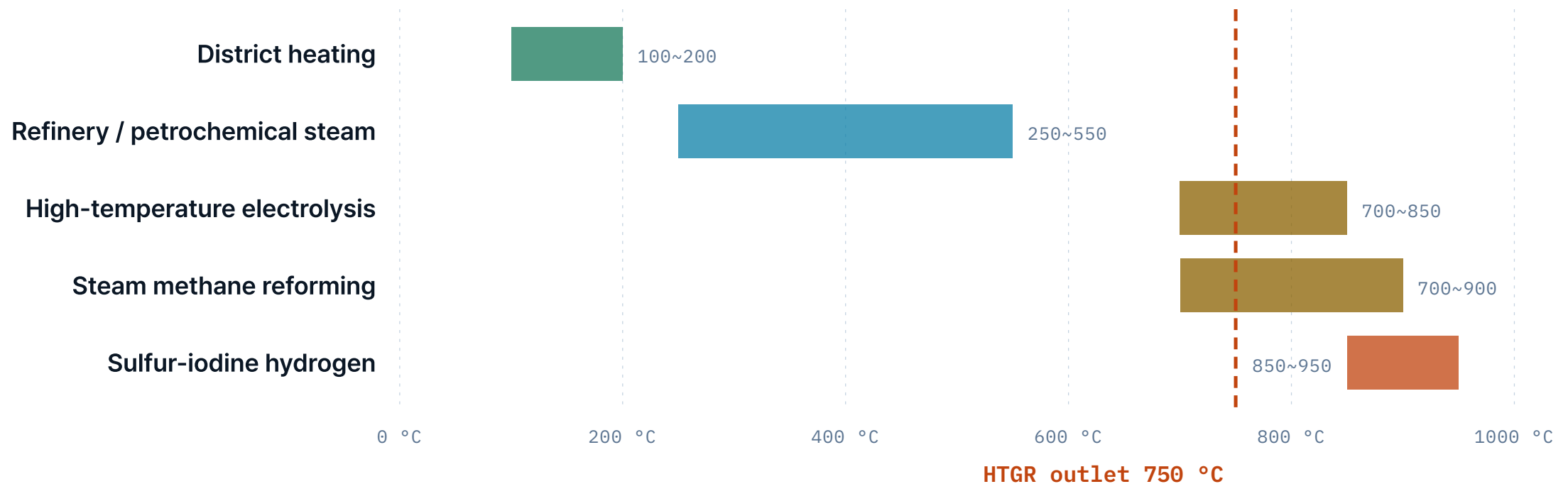
~116 MWe

equilibrium core fresh-fuel input and discharge in balance so the core composition is steady · **capacity factor** actual energy ÷ energy at rated power all the time · **first-of-a-kind** the first unit of a design

Source This textbook §4.5 · Chinese industry statement of May 2026 · cumulative capacity factor from the World Nuclear Association database

Temperatures that each use needs, against the 750 °C HTGR outlet

- The HTGR's real market is **heat**, not power - Xe-100's first customer is a chemical firm, BANR's is the army, Korea's partners are steel and petrochemicals
- 750 °C is the target because it **covers most of this list**. HTR-PM has supplied 12 MW of district heat since March 2024 - the first Gen IV reactor to heat a city



process heat heat used directly by industry rather than turned into power · **high-temperature electrolysis** splitting steam electrically at high temperature · **steam methane reforming** hydrogen from natural gas and steam · **sulfur-iodine (SI)** a thermochemical hydrogen cycle

Source This textbook §4.6 · HTR-PM joined the Rongcheng district-heating grid in March 2024 - 190,000 m², about 1,850 homes, 12 MW

05 HTGR safety

The light-water safety sentence is “keep supplying coolant”. The HTGR’s is “do nothing”. We read exactly how much of that the 2023 HTR-PM tests proved at commercial scale.

Three conditions for surviving without cooling

- With the coolant gone and the circulator stopped, if **conduction** → **radiation** alone keeps the fuel under its limit, the reactor does not need cooling
- Condition **2 is the crux** - the cap on module power is set by **core diameter**, not economics

LWR

"Keep supplying coolant"

ECCS, auxiliary feed, sprays - all variations on it

HTGR

"Do nothing"

graphite conduction → vessel radiation → cavity wall

1

Low power density

3.3 MW/m³ - little heat per unit volume

2

A slim core

short conduction path · 3 m across

3

Large heat capacity

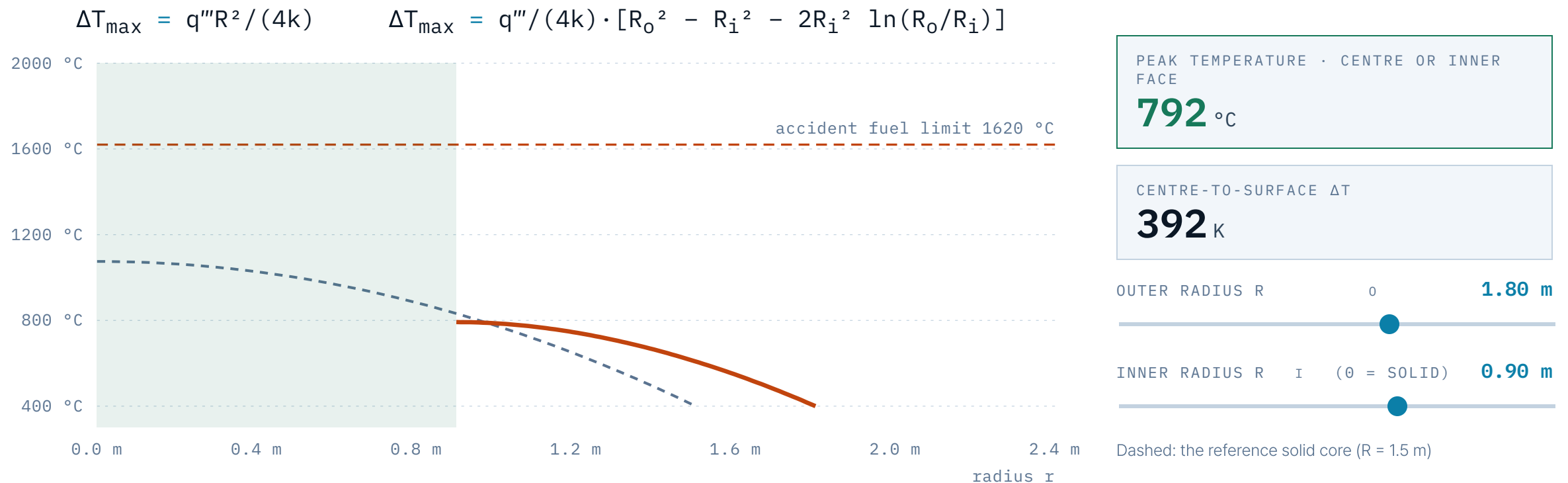
graphite soaks up the first hours

cavity the concrete room around the reactor vessel · **annular core** a ring-shaped core with a graphite column in the middle

Source This textbook §5.1 · heat conducts out through the graphite and radiates from the vessel surface to the cavity wall

Core temperature with conduction alone - solid and annular cores

- The centre-to-surface difference goes as R^2 - double the diameter, four times the difference. That is why 250 MWt cannot become 500 MWt by widening the core
- To grow, **hollow out the centre** - an annular core runs about 280 K cooler for the same heat (Xe-100 and others)

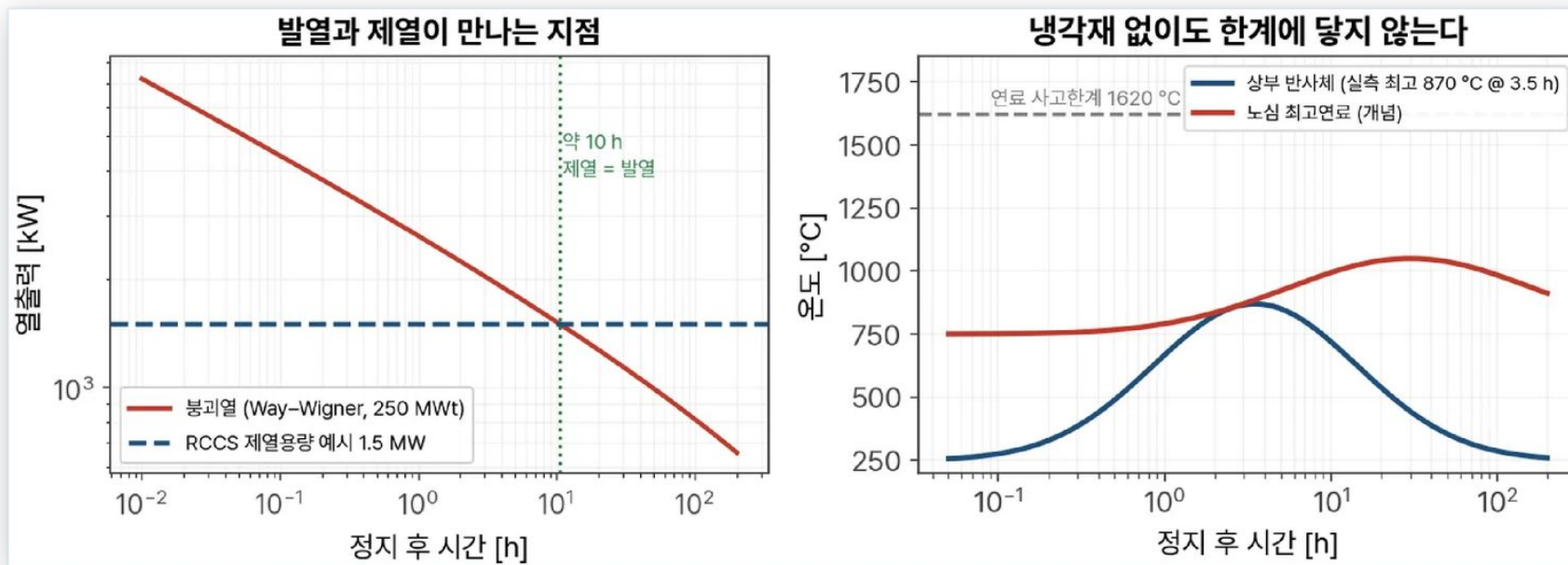


q''' heat generated per unit volume · k thermal conductivity · R_o, R_i outer and inner core radius · **adiabatic** no heat crosses the boundary

Calculation This textbook §5.2 · same conditions as Fig. 5.1 - decay heat $q''' = 30 \text{ kW/m}^3$, $k = 25 \text{ W/(m·K)}$, surface 400 °C · the annulus is adiabatic inside · accident fuel limit 1620 °C (HTR-PM)

When decay heat meets RCCS heat removal

- Decay heat is about 6 % at shutdown and 1 % after an hour. After 100 hours it is about 0.33 % - **about 810 kW** for a 250 MWt module, small enough to radiate from a few tens of m² of vessel
- Radiation goes as **T⁴** - the hotter the vessel, the faster it sheds heat, so the system stabilises itself. The worse it gets, the more it removes



Left - decay heat (250 MWt) meets an illustrative 1.5 MW RCCS at about 10 h (measured 13 h: next slide) · right - temperatures stay below the limit with no coolant · textbook Fig. 5.2

$$P_d/P_0 = 0.0622[t^{-0.2} - (t+10^8)^{-0.2}]$$

$$Q_{\text{rad}} = \epsilon_{\text{eff}} \sigma A (T_v^4 - T_c^4)$$

AT SHUTDOWN · AFTER 1 H

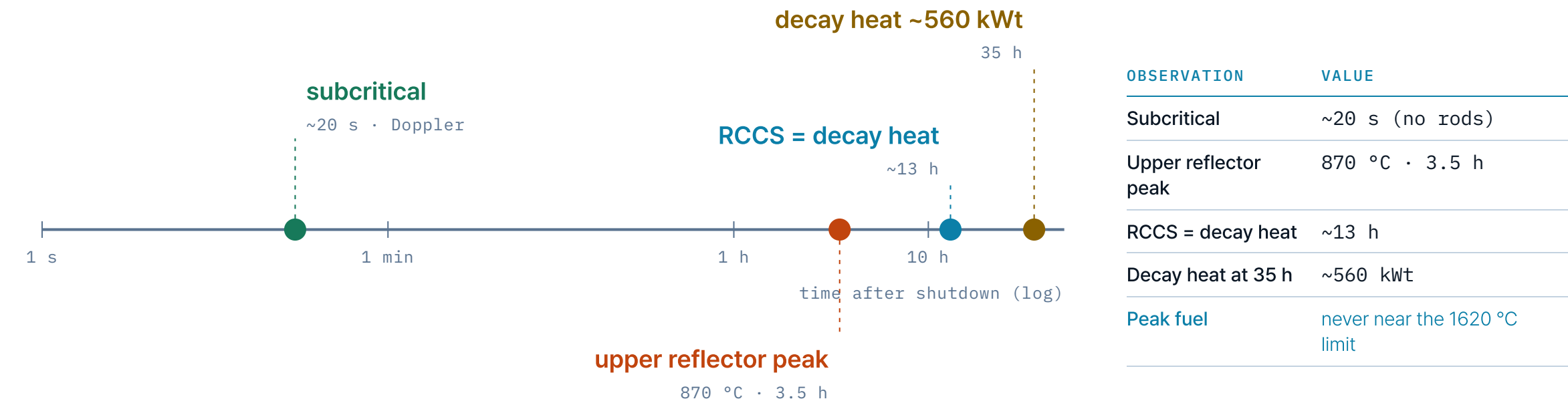
6 % · 1 %

RCCS Reactor Cavity Cooling System - air or water channels on the cavity wall, driven by the chimney effect · ϵ_{eff} effective emissivity · σ the Stefan-Boltzmann constant

Source This textbook §5.2, Fig. 5.2 · decay heat from Way-Wigner (t in seconds after shutdown) · the 1.5 MW RCCS is the figure's illustrative value · radiation goes as T⁴

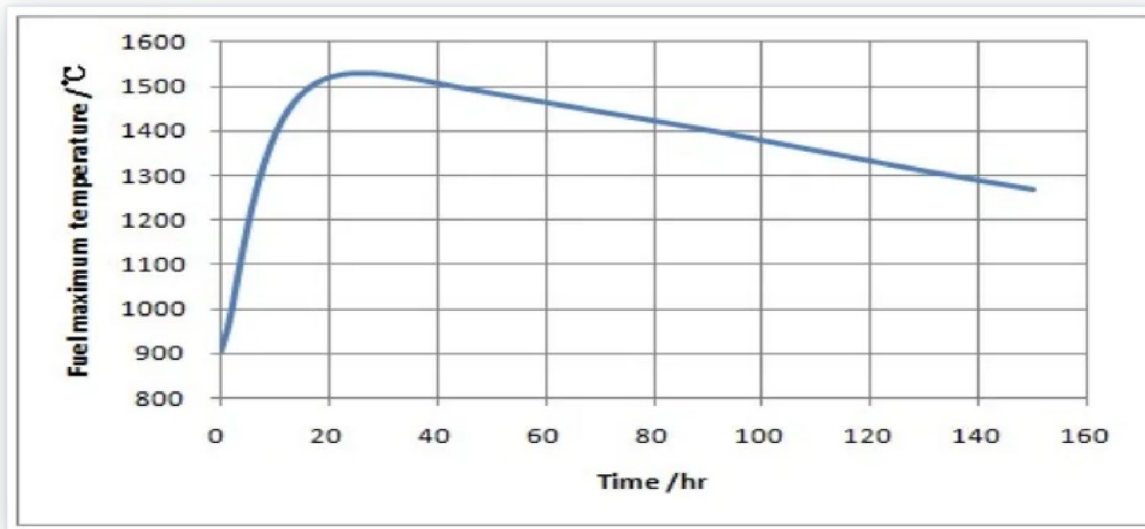
The 2023 HTR-PM loss-of-cooling demonstration - results

- The most important data in this book - **the only time loss of cooling has actually been done on a commercial-scale reactor**
- Both circulators stopped, with **no emergency core cooling and no powered cooling of any kind**

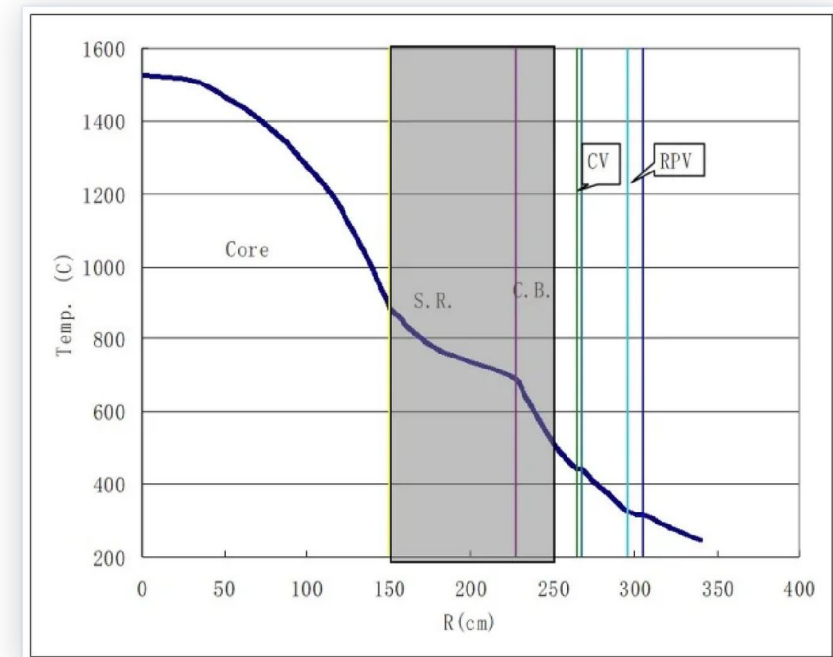


Three things to read from the demonstration

- ① Subcritical in 20 s by Doppler, not rods · ② peaks arrive late, 3.5 to 35 hours - minutes for an LWR, days here. Coping time is well beyond 24 hours
- ③ The peak is in the **upper reflector**, not the core - with flow stopped the heat below rises. Steady-state intuition gets accidents wrong



Calculated peak fuel temperature on loss of cooling - a peak in the 1500s °C near 25 h · Fig. 5.3



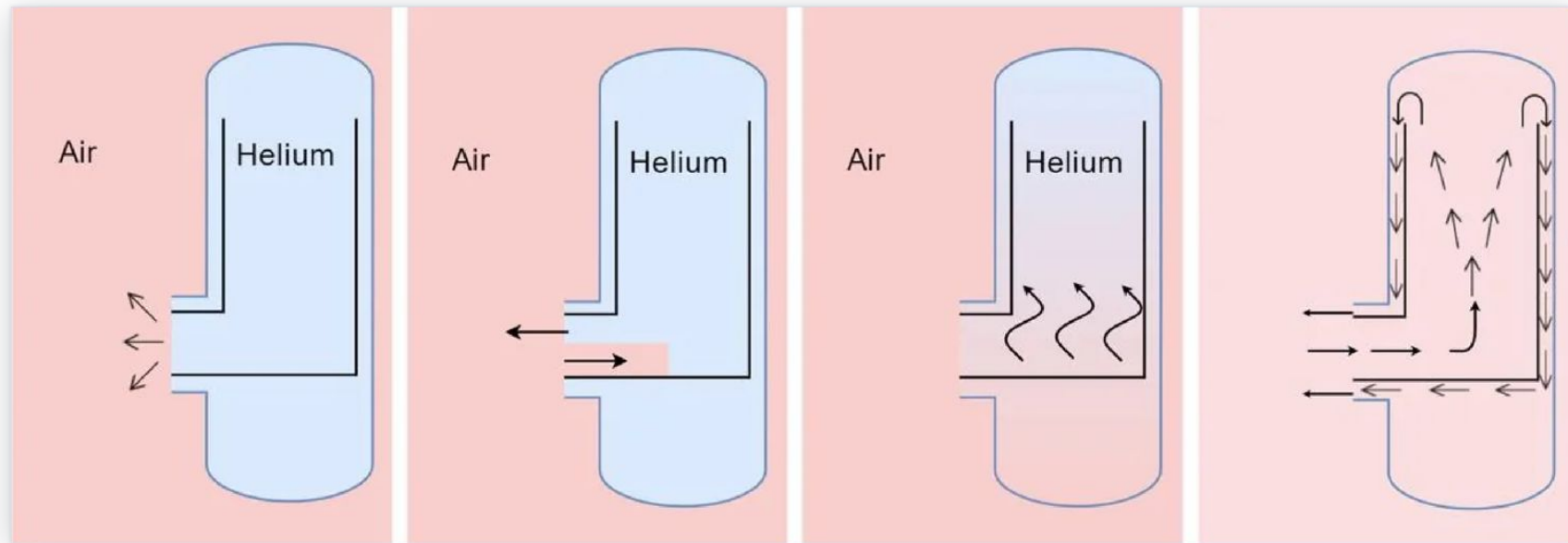
Accident radial temperatures - core out to the vessel · Fig. 5.4

Doppler effect hotter fuel broadens U-238 resonance absorption and cuts reactivity · pcm 10^{-5} in reactivity · \$ reactivity divided by the delayed fraction β · **coping time** how long the plant lasts without outside help

Source This textbook §5.3, Figs. 5.3-5.4 · coefficient about $-5 \times 10^{-5} / K \times 200 K = -1000$ pcm (about -1.5 \$) · S.R. side reflector, C.B. core barrel, CV containment, RPV reactor vessel

Air ingress - four stages and graphite oxidation

- The HTGR worries about **chemistry**, not temperature - the danger is not burning but **structural graphite thinning until it loses support**, and reaction heat raising local temperature
- ④ Natural circulation is ordinary system-code work. ② **Stratified counter-flow** and ③ **diffusion** cannot be computed by a single-phase homogeneous code - why GAMMA was born



SPECIES TO TRACK

He · N₂ · O₂ · CO · CO₂ ·
H₂O

Four stages of air ingress - ① inflow on depressurisation → ② stratified counter-flow → ③ molecular diffusion → ④ onset of natural circulation · textbook Fig. 5.5

stratified counter-flow heavy air flowing in low while light helium flows out high · **molecular diffusion** species mixing driven by concentration alone · **GAMMA** KAERI's multi-component gas code

Source This textbook §5.4, Fig. 5.5 · above 900 °C graphite oxidation only erodes the surface and is slow (§3.5)

Water ingress - three things at once

- A failed SG tube pushes **13 MPa water into 7 MPa helium** - one of HTR-PM's principal design-basis accidents
- It is easy to miss that **the reactivity sign is the opposite of an LWR's** - water, a moderator, raises reactivity in a graphite-moderated core

1

Pressure rise

Water flashes and primary pressure climbs. If a safety valve lifts, activity can escape.

2

Graphite-steam reaction

$C + H_2O \rightarrow CO + H_2$. Hydrogen is produced.

3

Reactivity insertion

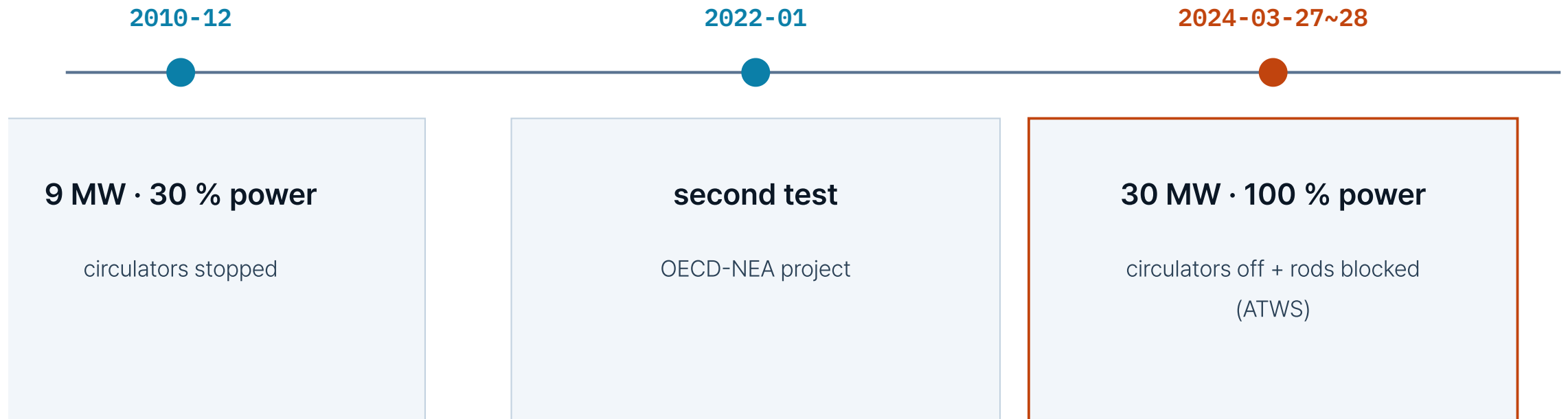
Water moderates. In a graphite-moderated thermal core it raises reactivity - the opposite sign to an LWR.

moderator a material that slows neutrons · **thermal reactor** one that fissions with slow neutrons · **safety valve** opens on overpressure to relieve it

Source This textbook §5.4 · steam generator secondary about 13 MPa against 7 MPa primary helium · HTR-PM splits the SG into 19 units with leak detection and isolation

The three HTTR **loss-of-forced-cooling** tests

- The 2024 test stopped the circulators at **full 30 MW** with **rod insertion blocked (ATWS)** - power fell and settled by itself, with no melting
- That both **pebble-bed (HTR-PM)** and **prismatic (HTTR)** have now been demonstrated matters a great deal for licensing



JAEA - "world-first full-power safety demonstration of a prismatic HTGR"

LOFC Loss of Forced Cooling · **ATWS** Anticipated Transient Without Scram - rods not inserted · **JAEA** Japan Atomic Energy Agency · **circulator** the gas blower

Source This textbook §5.5 · an OECD-NEA joint project extended to March 2027 · Korea participates

RCCS capacity, and where the HTGR sits in defence in depth

- The RCCS is air or water channels on the cavity wall - **driven by the chimney effect, with no pump, valve or power**
- The HTGR moves **level 3 (mitigation)** into the fuel grain and level 4 (severe accidents) into physics (conduction and radiation) - system reliability becomes **a materials and chemistry question**, and licensing follows

DESIGN	MEDIUM	NORMAL	ACCIDENT	
GA-MHTGR	natural-draught air	718 kW	1.75 MW	LEVEL 3 · MITIGATION → inside the fuel grain
Framatome SC-HTGR	water	1.4 MW	2.1 MW	
HTR-10	water	125 kW × 2 loops	—	LEVEL 4 · SEVERE ACCIDENTS → conduction · radiation

RCCS Reactor Cavity Cooling System · **chimney effect** warm air rising and driving its own flow · **defence in depth** independent layered protection · **GA-MHTGR** General Atomics’ modular HTGR

Source This textbook §5.6, Table 5.2 · the five-level defence-in-depth frame is from *SMR Design Requirements and Engineered Safety Features*

06 Running a system code on helium

We put helium into the control-volume code written in Volume III. Only two layers change - properties and closures. But the fact that graphite is the lead and helium the errand boy moves the centre of gravity of the analysis.

Which code layers **change** with the coolant, and which stay

- Only **layer 1 (properties)** and **layer 2 (closures)** change; numerics and plant layout stay - the strength of a control-volume code
- "Stay" has a footnote - **time-step control behaves differently with each fluid** (Courant, below)

	물성 클래스	전열·마찰 상관식	시간간격·수치	추가 물리
VHTR (He)	이상기체 $P = \rho RT$	KTA 3102.2/.3	음속 1880 m/s 음해법 필수	고체 전도·복사 다성분 확산
SFR (Na)	비압축 액체 $\rho(T)$ 선형	Mikityuk $Nu(Pr)$	저압·큰 열관성 긴 적분시간	축방향 전도 풀 성층화
MSR (염)	염 물성표 (불확실 10–50 %)	물 기준 상관식 재평가 필요	동결 판정 $T > T_{melt}$	선구핵종 수송 유체 내 발열

노심·연결 구조는 그대로 두고, 이 네 칸만 갈아끼운다

Swap map by reactor type - keep the core connectivity, change four boxes · textbook Fig. 6.1

LAYER 1 · PROPERTIES
p, h → ρ , T, derivatives

LAYER 2 · CLOSURES
**friction · heat transfer ·
choking**

LAYER 3 · NUMERICS
**discretisation · pressure eq. ·
 Δt**

LAYER 4 · PLANT
how volumes are joined

control volume a finite volume for mass and energy accounting · **junction** the flow path between volumes · **closure** correlations such as friction and heat transfer that close the equations · **semi-implicit** pressure implicit, the rest explicit

Source This textbook §6.1, Fig. 6.1 · the four-layer structure of *Writing a Reactor System Code*, Volume III

The helium property class - closed forms and the speed of sound

- Derivatives once taken numerically from IAPWS tables now come in **closed form** - nothing leaves a table or chatters across saturation, so it is **far more robust than water**
- A rare case where the pressure equation's diagonal can be shown by hand to **equal** the physical speed of sound - derive it yourself

$$T = h/c_p, \quad \rho = p/(RT)$$

$$\rho_p = 1/(RT), \quad \rho_h = -p/(RT^2 c_p)$$

$$\rho_p + \rho_h/\rho = 1/(RT) - 1/(T c_p) = (1/RT)(1 - R/c_p) = (1/RT)(c_v/c_p)$$

$$c = 1/\sqrt{\rho_p + \rho_h/\rho} = \sqrt{\gamma RT}$$

SPEED OF SOUND · FROM THE PRESSURE EQ.

1,882 m/s

$\sqrt{\gamma RT}$

1,882 m/s

HELIUM TEMPERATURE

750 °C

ρ_p , ρ_h partial derivatives of density with pressure and with enthalpy · **IAPWS** the international water and steam property standard · **diagonal term** the diagonal of the pressure-equation matrix

Source This textbook §6.2 · putting an ideal gas into Volume III's $c = 1/\sqrt{\rho_p + \rho_h/\rho}$ gives $\sqrt{\gamma RT} \cdot R = 2077.3$, $c_p = 5195$ J/(kg·K)

A six-volume model of one HTR-PM module, and the heat-capacity ratio

- Graphite is the lead, helium the errand boy - the solid holds 73 times the fluid's heat. HeatStructure nodes and irradiated graphite conductivity dominate the answer
- Two time scales - seconds for the gas, hours to days for the solid (ratio $\sim 10^4$, stiff). In practice **blowdown and conduction cooling are solved separately**



total 146 m³ · about 640 kg of helium · a loop transit of about 6.7 s at 96 kg/s

Helium (fluid)

3.3 MJ/K

Core graphite (solid)

240 MJ/K

the solid holds 73 ×

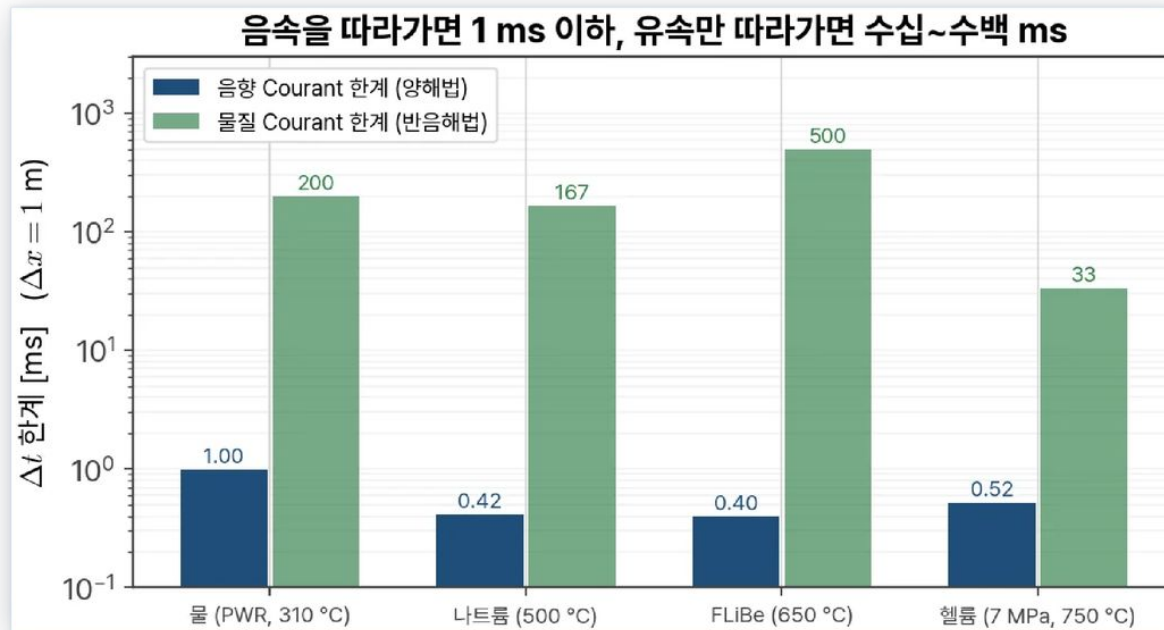
HeatStructure the code object that solves conduction in solids · **stiff** a system mixing very different time scales, forcing tiny steps ·

blowdown depressurisation after a break

Calculation This textbook §6.3, htgr_loop.py · volumes total 146 m³ · helium about 4.4 kg/m³ × 146 m³ ≈ 640 kg · 640 × 5195 = 3.3 MJ/K · graphite 78 m³ × 1750 kg/m³ × 1759 J/(kg·K) = 240 MJ/K · loop transit 640/96 ≈ 6.7 s

The two Courant limits on the time step, and helium's velocity

- What sets helium apart is **velocity**, not sound speed - low density means about 30 m/s, six times water → even semi-implicit, **the material Courant limit is six times tighter**
- Volume III's IdealGas defaults $\mu = 3.9 \times 10^{-5}$, $k = 0.29$ are **~450 °C values**; at 750 °C they are 4.7×10^{-5} and 0.37 - **left constant, the outlet is off by tens of kelvin**



$$\Delta t_{\text{sonic}} \leq \Delta x / (u + a)$$

$$\Delta t_{\text{material}} \leq \Delta x / u$$

HELIUM, SEMI-IMPLICIT (7 MPa, 750 °C)

33 ms

WATER, SEMI-IMPLICIT (PWR, 310 °C)

200 ms

Time-step limits by coolant ($\Delta x = 1$ m) - explicit never passes 1 ms; semi-implicit reaches tens to hundreds of ms · textbook Fig. 6.2

Courant condition the stability rule that information may not cross more than one cell per step · **explicit** next value from current values only · **semi-implicit** pressure solved implicitly, removing the sonic limit

Source This textbook §6.4, Fig. 6.2 · $\Delta x = 1$ m · helium at 750 °C carries sound at 1882 m/s (water about 1000 m/s) · velocity about 30 m/s (about 6× water)

What the Volume III code **cannot** solve

- Normal operation, loss of flow and depressurisation run. But each volume holds **one fluid** - air ingress needs per-species mass equations and diffusion, so **half the code would be rewritten**
- A 1-D chain misses the bypass along the wall, where porosity climbs **from 0.39 to nearly 1.0**

$$\partial(\rho Y_i)/\partial t + \nabla \cdot (\rho u Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i$$

Solves

Normal operation

single-phase helium, forced flow

Loss of flow

circulator trip, into conduction

Depressurisation

one fluid leaving

Does not

Air ingress

per-species equations, diffusion

Pebble-bed wall bypass

porosity 0.39 → 1.0, multi-D

Oxidation heat

the reaction source S_i

KAERI's **GAMMA / GAMMA+** code was built for exactly this. The lab already has a primer - **go there** rather than re-deriving it here.

multi-component mixture several gas species, each needing its own conservation equation · **porosity** void fraction · **bypass** flow slipping along the wall past the core · **porous-medium model** treating the bed as a sponge of average porosity

Source This textbook §6.5 · Y_i mass fraction of species i , D_i its diffusivity, S_i its reaction source · continue with the lab's GAMMA+ primer

Integral data for validating HTGR system codes

- Harder than writing the code is **checking that it is right**. The integral data for HTGRs are the four rows below - nothing more
- The last row changed the field - **the HTGR is the only non-LWR with commercial-scale integral data**. Know this before designing a code project

DATA	SCALE	WHAT IT VALIDATES
HTTR LOFC tests (2010, 2022, 2024)	30 MWt	prismatic conduction cooling, ATWS
HTR-10 circulator trip without scram	3 MWt	pebble-bed temperature feedback
NACOK facility (Germany)	non-nuclear	air-ingress natural circulation
HTR-PM loss-of-cooling tests (2023)	200 MWt	full conduction cooling at commercial scale

Six numbers to keep from Part 1

- What helium cannot carry, graphite stores; while it stores it, TRISO holds the fission products - **these six numbers are the evidence**

7 MPa

pressure to make blower power affordable

$\approx 4900 \times$ the power at 1 atm

3.3 MW/m³

passive safety is bought with volume

a 78 m³ core vs 2.5 m³ for an LWR

1600 °C

just before the TRISO curve bends

+200 K, 15 $\times$ the failures

$\Delta T \propto R^2$

module power is capped by diameter

grow it with an annular core

20 s · 35 h

the 2023 HTR-PM demonstration

Doppler shutdown · peaks a day later

73 $\times$

graphite leads, helium runs errands

240 vs 3.3 MJ/K, solid vs gas

Abbreviations (1 of 2) - reactors, fuel, materials

Abbreviation	Expansion	What it means
HTGR	High Temperature Gas-cooled Reactor	all helium-cooled, graphite-moderated reactors
VHTR	Very High Temperature Reactor	Gen IV class aiming at 900 °C and up
FHR	Fluoride-salt-cooled High-temp. Reactor	solid TRISO fuel, fluoride-salt coolant
MSR · SFR	Molten Salt / Sodium-cooled Fast Reactor	the next two decks
HTR-PM	High Temperature Reactor – Pebble-bed Module	the commercial HTGR at Shidaowan
HTTR	High Temperature engineering Test Reactor	Japan's 30 MWt prismatic test reactor
TRISO	Tri-structural Isotropic particle	coated particle - each one a containment
UCO	Uranium Oxycarbide	oxide-carbide kernel that suppresses CO
SiC · PyC	Silicon Carbide · Pyrolytic Carbon	the pressure boundary and carbon layers
HALEU	High-Assay Low-Enriched Uranium	uranium enriched to 5-20 %
FCM	Fully Ceramic Microencapsulated fuel	TRISO in a ceramic matrix (MMR)
FIMA	Fissions per Initial Metal Atom	burnup as a fraction of initial metal
dpa	Displacements per Atom	radiation dose to graphite
GWd/tU	Gigawatt-days per tonne of U	burnup per tonne of uranium
IG-110 · NBG-18	—	nuclear graphite grades for HTGRs

Abbreviations (2 of 2) - systems, tests, safety, organisations

Abbreviation	Expansion	What it means
RCCS	Reactor Cavity Cooling System	cavity cooling by the chimney effect, unpowered
AMB	Active Magnetic Bearing	no oil in the circulator
SAS	Small Absorber Sphere system	absorber balls dropped into reflector holes
LOCA · LOFC	Loss of Coolant · of Forced Cooling	loss of coolant, loss of forced cooling
ATWS	Anticipated Transient Without Scram	a transient with rods not inserted
AGR	Advanced Gas Reactor fuel programme	the US TRISO fuel test programme
ATR · FACS	Advanced Test Reactor · furnace	INL test reactor, post-irradiation furnace
EFPD	Effective Full-Power Days	effective full-power days
KTA	Kerntechnischer Ausschuss	German standards - pebble-bed correlations
GAMMA+	—	KAERI's multi-component gas code
NACOK	—	German air-ingress test facility
INET	Institute of Nuclear and New Energy Tech.	Tsinghua institute that designed HTR-PM
JAEA	Japan Atomic Energy Agency	Japan's agency that runs HTTR
FONSI	Finding of No Significant Impact	US environmental finding
MMR · BANR	Micro Modular / BWXT Adv. Nuclear Reactor	two micro-HTGR designs

Passive safety is not free - it is bought with volume

Helium does nothing, graphite stores, TRISO holds.

The price is a 3.3 MW/m³ core and a 7 MPa circulator.

Next is the **SFR** - a reactor that **raises** power density instead. Why use a coolant that explodes with water?